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Atypical collective oscillatory activity in cardiac tissue uncovered by optogenetics

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This **important** work provides mechanistic insights into the development of cardiac arrhythmia and establishes a new experimental use case for optogenetics in studying cardiac electrophysiology. The agreement between computational models and experimental observations provides a **convincing** level of evidence that wave train-induced pacemaker activity can originate in continuously depolarized tissue, with the limitation that there may be differences between depolarization arising from constant optogenetic stimulation, as opposed to pathophysiological tissue depolarization. Future experiments in vivo and in other tissue preparations would extend the generality of these findings.

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Abstract

Many biological processes emerge as frequency-dependent responses to trains of external stimuli. Heart rhythm disturbances, i.e. cardiac arrhythmias, are important examples as they are often triggered by specific patterns of preceding stimuli. In this study, we investigated how ectopic arrhythmias can be induced by external stimuli in cardiac tissue containing a localised area of depolarisation. Using optogenetic *in vitro* experiments and *in silico* modelling, we systematically explored the dynamics of these arrhythmias, which are characterized by local oscillatory activity, by gradually altering the degree of depolarization in a predefined region. Our findings reveal a bi-stable system, in which transitions between oscillatory ectopic activity and a quiescent state can be precisely controlled, i.e. by adjusting the number and frequency of propagating waves through the depolarized area oscillations could be turned on or off. These frequency-dependent responses arise from collective mechanisms involving stable, non-self-oscillatory cells, contrasting with the typical role of self-oscillations in individual units within biophysical systems. To further generalize these findings, we demonstrated similar frequency selectivity and bi-stability in a simplified reaction-diffusion model. This suggests that complex ionic cell dynamics are not required to

reproduce these effects; rather, simpler non-linear systems can replicate similar behaviour, potentially extending beyond the cardiac context.

Introduction

Biological systems like the heart typically display complex dynamical behaviour through interactions among components such as genes, proteins, or cells (Dana et al. (2008)). These systems exhibit non-linear dynamics, feedback loops, and adaptability, reflecting phenomena like homeostasis, evolution, and rhythm stability (Xiong and Garfinkel (2023)). Central to the framework of dynamical systems is the concept of the phase space, a multidimensional representation in which each point corresponds to a unique state of the system. In geography when a small ball rolls on a raised-relief map, a state would include the planar position, elevation, speed and acceleration of the ball.

Within the phase space, the long-term behaviour of a system can often be described in terms of attractors, which are subsets of the phase space that trajectories tend toward over time. Attractors can take various forms, including fixed points, periodic orbits, and chaotic sets, reflecting the rich diversity of behaviours possible in dynamical systems. By replacing the ball by local rain in the raised-relief map, rivers, lakes and other bodies of water are attractors where that rain is collected. A watershed, also known as a drainage basin, is an area of land where any rain that falls will drain into the same river, lake or body of water. Therefore they are basins of attraction. Mountain ridges are separatrices that divide the watersheds, or more formally theoretical surfaces that divide the phase space into different basins of attraction. A separatrix therefore denotes the transition between different regions of the phase space. The terminology introduced here comes back throughout our paper, with the addition of a “state”, i.e. one of the attractors in the system. When the system is not interacted with, it will stay at the same point(s) in phase space.

Some systems only have a single basin of attraction. However, in others the phenomenon of bi-stability emerges, where two distinct attractors coexist, each with its own basin of attraction. Bistability is a hallmark of many natural and engineered systems, from climate dynamics to neural networks, ecological models, and cardiac dynamics (Angeli et al. (2004)). The interplay between basins in such systems can give rise to complex behaviours, making the characterization of these basins a critical component of a system’s analysis.

In this manuscript, the focus is on the complex system of the heart. At rest, ventricular cardiomyocytes (i.e muscle cells of the lower heart chamber) have a membrane potential of -80 to -90 mV. This resting state is an attractor of the system. When pushed out of equilibrium (e.g. due to an influx of Na^+ ions or electrical stimulation), cardiomyocytes first depolarize and subsequently get attracted again towards the resting state by an efflux of K^+ ions. In phase space the trajectory that gets mapped out resembles a cycle, corresponding to one heartbeat. When multiple of these cells are connected to each other, emergent behaviour can arise. Within this dynamic system of cardiomyocytes, we investigated emergent bi-stability (a concept that will be explained more thoroughly later on) in cell monolayers under the influence of spatial depolarization patterns. By using a train of external (electrical) waves, the system could be actively moved from one basin of attraction to another.

Cardiac arrhythmias are typically initiated by specific triggers, with point-source waves, i.e. ectopic waves being the most prevalent. To study ectopy in laboratory settings, control needs to be exerted over the depolarisation of cardiomyocytes. This can be achieved by optogenetics, i.e. a biological technique to control the activity of cells with light. This approach relies on the introduction of recombinant genes encoding light-activatable proteins into cells to endow them with a new biological function (Tan et al. (2022)). One way to depolarise a cardiomyocyte is by using mini-singlet oxygen generator (miniSOG). Upon activation with blue light, miniSOG converts molecular oxygen from its ground state ($^3O_2$) into highly reactive singlet oxygen ($^1O_2$). The resulting oxidative damage induces a state of prolonged depolarisation in cardiomyocytes following electrical stimulation (Jangsangthong et al. (2016)). In such a system, it was previously observed that spatiotemporal illumination can give rise to collective behaviour and ectopic waves (Teplenin et al. (2018)) originating from illuminated/depolarised regions (with spatial high curvature). These waves resulted from the interplay between the diffusion current (also known in biology/biophysics as the gap junction mediated current) and the single cell bi-stable state (Teplenin et al. (2018)) that was induced in the illuminated region.

Although effects of illuminating miniSOG with light might lead to formation of depolarised areas, it is difficult to control the process precisely since it depolarises cardiomyocytes indirectly. Therefore, in this manuscript, we used light-sensitive ion channels to obtain more refined control over cardiomyocyte depolarisation. These ion channels allow the cells to respond to specific wavelengths of light, facilitating direct depolarization (Ördög et al. (2021, 2023)). By inducing cardiomyocyte depolarisation or hyperpolarisation only in the illuminated areas, optogenetics enables precise spatiotemporal control of cardiac excitability, an attribute we exploit in this manuscript (**Appendix 2—figure 1**). With the aid of the light-gated ion channel CheRiff (Hochbaum et al. (2014)), ectopic waves similar to those obtained in the miniSOG system could be elicited (**Appendix 2—figure 2**). Due to the superior control over depolarisation imposed by CheRiff, we uncovered new effects and unobserved behaviour related to the transition from a mono-stable cardiac system to a bi-stable one that displays periodic oscillations. This transition resulted from the interaction of depolarized cardiomyocytes in an illuminated region with an external wave train not originating from within the illuminated region. More specifically, we found that our system displayed frequency selectivity: it oscillated only as a reaction to a specific range of inter-pulse external stimuli times. In neuroscience, this behaviour is classified as “resonance”. However, to avoid confusion with classical resonance, where the strength of the response is dependent on the input, here the phenomenon is called “induced pacemaker activity”. Evidence for this induced pacemaker activity will be presented both *in vitro* and *in silico*.

Results

Bi-stability in cardiomyocyte monolayers was realised both *in vitro* and *in silico* (**Figure 1**). *In vitro* this happened through patterned illumination applied to neonatal rat ventricular myocyte (NRVM) monolayers, which were optogenetically modified using the blue light-activatable cation channel CheRiff (**Appendix 2—figure 1**). *In silico*, use was made of a slightly modified (see Methods and Materials) NRVM computational model (Majumder et al. (2016)) complemented with a model for a depolarising light-gated ion channel (Williams et al. (2013)). In both cases, the illumination pattern consisted of a rectangle of variable light intensities. After initiating a periodic wave train (of variable pulse number and interpulse interval/period) from a pacing electrode, two main outcomes and one special case were observed (**Figure 1A**). The first outcome was that the tissue stayed in the stationary rest state (STA). The second outcome was that the illuminated region became a pacemaker, which sent out pulses from its edges as a form of oscillatory behaviour (OSC). The special case consisted of an in-between case, in which the illuminated region only sends out a limited amount of pulses before returning to the stationary state thus showing transient oscillatory behaviour (TR. OSC). Except for the occasional mention, this last behaviour of transient oscillations will be reserved for more detailed discussion in the supplement. All light intensity-influenced monolayer behaviours are visualised together in **Appendix 2—figure 3** and the accompanying Supplemental video 1. Here, we will focus on the two main behaviours after periodic wave trains: STA and OSC. As shown below, these two behaviours can exist within the same monolayer under bi-stability and transitions from one state to the other are possible using periodic wave trains. The occurrence of this phenomenon appears to be influenced by the light intensity applied to the illuminated region (**Figure 1B**). In a step-by-step manner, we unravelled the intricacies of this tissue level (i.e. collective) bi-stability between oscillatory and stationary behaviour, linking experiment to theory.

In vitro realisation of collective light-induced bi-stability

To test the scope of collective bi-stability *in vitro*, multiple parameters were varied within our experimental set-up. The first one of these was the frequency of stimulation, visualised in **Figure 2**, where the number of electrical pulses and the light intensity (in the range 0.03125-0.25 mW/mm^2) were kept fixed. In every NRVM monolayer (26 biological replicates/monolayers), three frequencies (5.000, 1.666, and 0.833 Hz or stimulation periods 200, 600, and 1200 ms) were tested subsequently, with a total of three repetitions (technical replicates) of the full set, totalling 26 × 3 = 78 observations at each pacing frequency. Optical voltage signals were measured both at the center of the illuminated area (top traces) as well as in the bulk of the tissue (bottom traces), in order to visualise action potentials throughout the tissue. Underneath these optical voltage traces is a space-time plot of the experimental monolayer, the vertical axis showing space and the horizontal one showing time. At each moment in time, the middle line of the monolayer image was plotted. White shows depolarized tissue, while black signifies repolarised cells. From these plots, intermediate stimulation periods (600 ms) were observed to change the monolayer from a collective stationary state to a collective oscillatory state, while low (200 ms) or high (1200 ms) ones had no effect. It was observed that the oscillatory state consisted of ectopic pacemaker activity with a frequency (1.361 Hz or a 735-ms period) that was different from the stimulus train frequency (1.666 Hz or a 600-ms period). Similar results were found for the transient oscillatory state (**Appendix 2—figure 4A**). In short, we found that collective bi-stability becomes apparent in the mid-range of electrical pacing frequencies.

A second parameter tested was the number of pulses, visualised in **Figure 3**, where the period of stimulation (600 ms) and the light intensity (in the range 0.03125-0.25 mW/mm^2) were kept fixed. In every NRVM monolayer (7 biological replicates/monolayers), two different pulse numbers (2 and 4) were tested subsequently, with a total of three repetitions (technical replicates) of the full set, totalling 7 × 3 = 21 observations for each number of pulses. Once again optical voltage traces were taken from the center of the illuminated area and from the bulk of the tissue. Our observations show that a minimum number of pulses is needed to change the collective behaviour of the monolayer from stationary to oscillatory.

Taken together, we found a way to control transitions of collective behaviour of cardiomyocyte monolayers from stationary to oscillatory through two different parameters, i.e. the frequency and number of pulses.

In silico realisation of collective light-induced bi-stability

Using the power of state-of-the-art computational modelling, we performed similar experiments to the *in vitro* ones with a modified version of the Majumder et al. (2016) model for NRVMs combined with the Williams et al. (2013) model for the H134R mutant of *Chlamydomonas reinhardtii* channelrhodopsin-2. As this light-sensitive ion channel slightly differs from CheRiff in its biophysical properties, the *in vitro* findings will have a qualitative rather than a quantitative correspondence to the *in silico* results, which are visualised in **Figure 4**. Simulating at a fixed constant illumination (1.7124 mW/mm^2) and a fixed number of 4 pulses, frequency dependency of collective bi-stability was reproduced in **Figure 4A**. Similar to the experimental observations, only intermediate electrical pacing frequencies (2.000 Hz or a 500-ms period) caused transitions from collective stationary behaviour to collective oscillatory behaviour and ectopic pacemaker activity had periods (710 ms) that were different from the stimulation train period (500 ms). **Figure 4B** shows the accumulation of pulses necessary to invoke a transition from the collective stationary state to the collective oscillatory state at a fixed stimulation period (600 ms). Also in the

Figure 1.

Bi-stability between stationary behaviour and collective oscillations.

(A) Experimental timeline for a monolayer of optogenetically modified NRVMs under constant illumination of its center: 1) observation of a collective stationary state (STA); 2) periodic wave train from the pacing electrode; 3) outcome observation of collective behaviour, either stationary (STA) or oscillatory (OSC). **(B)** Schematic representation showing how light intensity influences collective behaviour of excitable systems, transitioning between a stationary state (STA) at low illumination intensities and an oscillatory state (OSC) at high illumination intensities. Bi-stability occurs at intermediate light intensities, where transitions between states are dependent on periodic wave train properties. TR. OSC, transient oscillations.

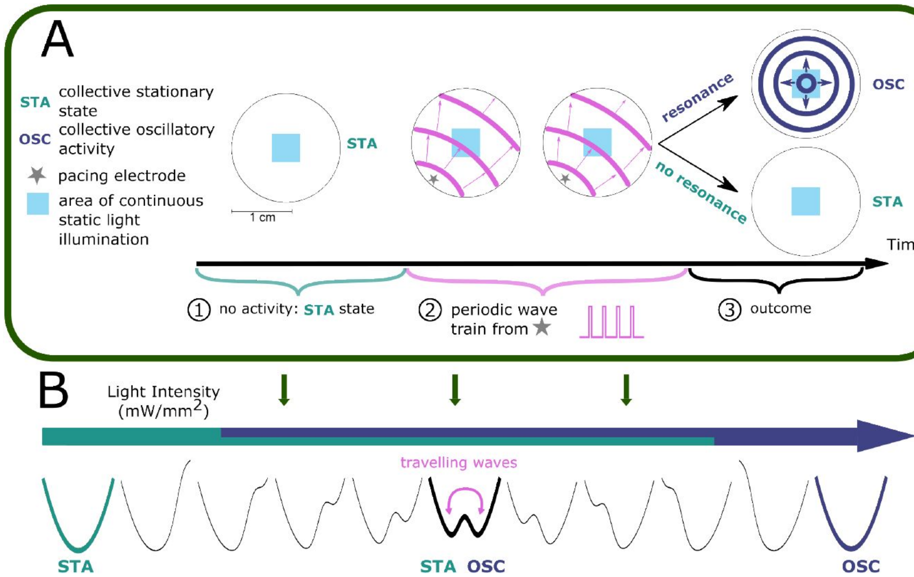
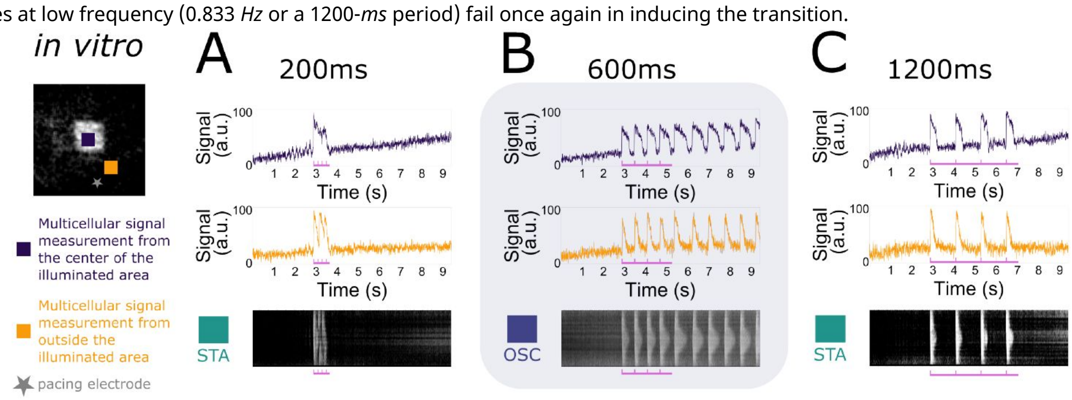


Figure 2.

Frequency dependency of induced collective pacemaker activity.

In vitro monolayers of optogenetically modified NRVMs show that the transition from the stationary state (STA) towards the oscillatory state (OSC) is dependent on the frequency of excitatory waves passing through an illuminated area under a constant number of 4 pulses. **(A)** Four pulses at high frequency (5.000 *Hz* or a 200-*ms* period) are not enough to induce the transition. **(B)** Four pulses at medium frequency (1.666 *Hz* or a 600-*ms* period) are sufficient to induce the transition. **(C)** Four pulses at low frequency (0.833 *Hz* or a 1200-*ms* period) fail once again in inducing the transition.



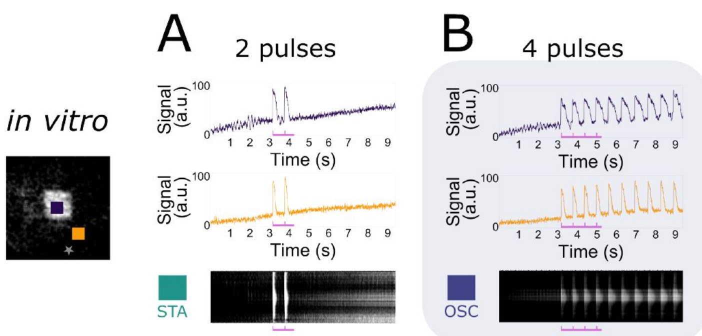


Figure 3.

Accumulation of pulses to induce collective pacemaker activity.

In vitro monolayers of optogenetically modified NRVMs show that the transition from the stationary state (STA) towards the oscillatory state (OSC) is dependent on the number of excitatory waves passing through an illuminated area at a constant electrical pacing frequency of 1.666 *Hz* or a 600-*ms* period. **(A)** Two pulses are not enough to induce the transition. **(B)** Four pulses are sufficient to induce the transition.

in silico simulations, ectopic pacemaker activity had periods (750 *ms*) that were different from the stimulation train period (600 *ms*). Also for the transient oscillatory state, the simulations show frequency selectivity (**Appendix 2—figure 4B**).

Having established a qualitative correspondence between the *in silico* model and *in vitro* experiments, the unique properties of the computational model were exploited to assert control over all variables in order to determine the parameter ranges of the state transitions. We previously identified three important parameters influencing such transitions: light intensity, number of pulses, and frequency of pulses. Scanning through all three variables revealed the specific conditions at which transition from the stationary state towards the oscillatory state occurred as shown in the rightmost panel of **Figure 4C** with slices for fixed illumination extracted from this 3D-plot shown at the left. With increasing light intensity, the parameter range (pulse and period) in which bi-stability can be detected increases (the total area of TR.OSC. + OSC. becomes larger), and transient oscillatory states transition towards oscillatory states.

Except for the three aforementioned parameters, a fourth one was found to have an influence on the state transitions: the size of the illuminated region. Also this parameter was first tested *in vitro* resulting in 5 biological replicates/monolayers showing transitions to the oscillatory state and 6 biological replicates/monolayers showing transient oscillations. The effect of the size of the illuminated area on the occurrence of bi-stability was qualitatively reproduced *in silico* (**Appendix 2—figure 5**). Varying the number and frequency of pulses together with the illuminated area at a fixed light intensity of 1.7200 mW/mm^2 resulted in the dynamic behaviour shown in **Figure 4D**. For the conditions that resulted in stable oscillations, the green vertical lines in the middle and right slices represent the natural pacemaker frequency in the oscillatory state. After the transition from the stationary towards the oscillatory state, oscillatory pulses emerging from the illuminated region gradually dampen and stabilize at this period, corresponding to the natural pacemaker frequency. The location of this line depends on the light intensity and the size of the illuminated region. However, it is independent of the number of pulses and the frequency of the stimulus train. **Figure 4C** and **Figure 4D** also illustrate that no oscillatory state could be induced with a high number of pulses for stimulus trains of high frequency (i.e. low stimulation period).

Termination of collective pacemaker activity

A closer look at the high frequency range showed that it not only prevented the initiation of ectopic pacemaker activity, but can also cause its termination (**Figure 5A**). In NRVM monolayers (3 biological replicates/monolayers), collective ectopic pacemaker activity was first initiated using a stimulus train of 3 pulses with a period of 500 *ms*. Once ongoing, 8 pulses in the high frequency range (5.000 *Hz* or period 200 *ms*) terminated the ectopic pacemaker activity and made the tissue go back to its resting state. When copying this experimental protocol *in silico*, qualitative correspondence was shown using 4 pulses with a period of 500 *ms* to initiate pacemaker activity, and 7 pulses with a period of 180 *ms* to terminate it. Note that both experimentally and computationally every pulse of the stimulus train is captured (orange traces), despite the fact that the voltage traces from points inside the illuminated region do not give that impression.

Once again, the qualitative *in silico* correspondence allows us to determine the specific conditions at which initiation and termination of collective ectopic pacemaker activity occur in more detail (**Figure 5B**). In the period vs. number of pulses plot, which is a copy of the rightmost slice in **Figure 4C**, the region showing termination (red-lined) did not overlap with the region showing initiation (magenta-lined).

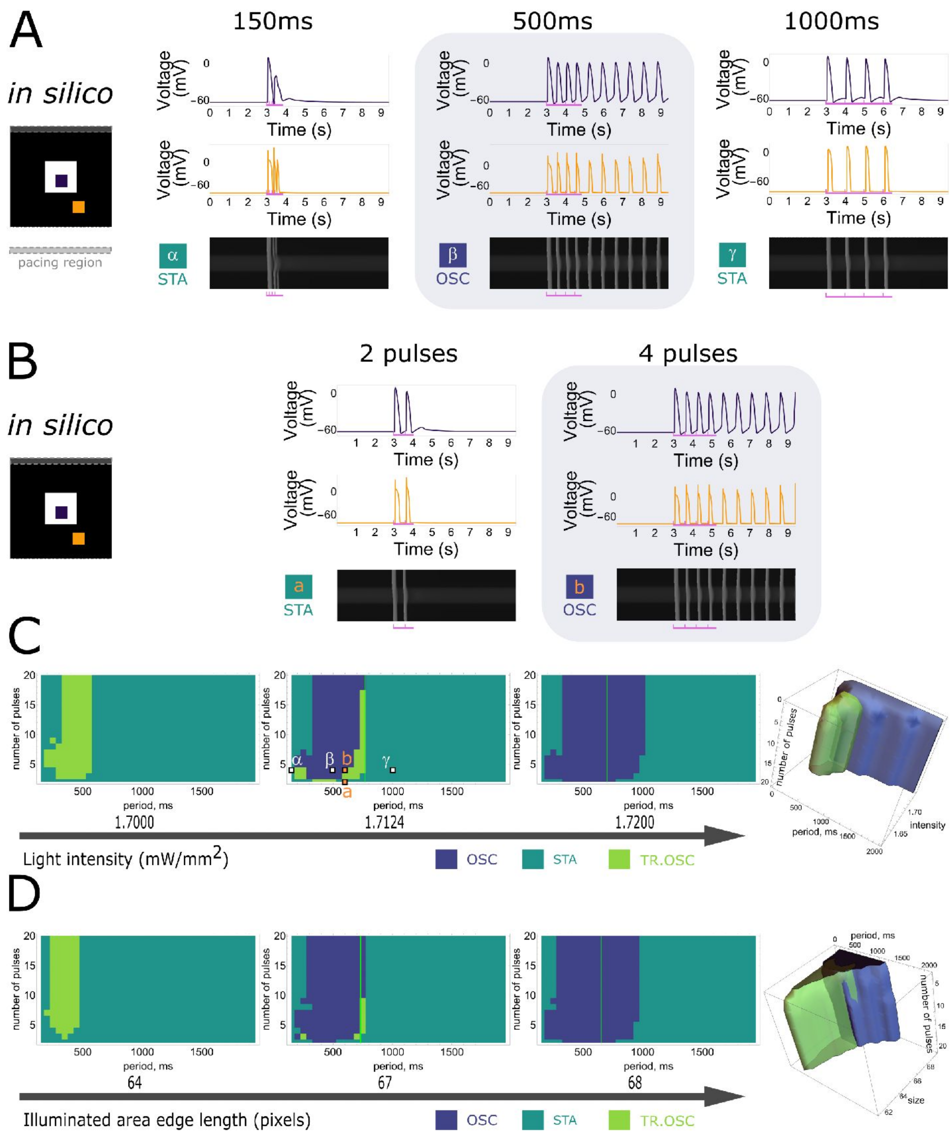


Figure 4.

***In silico* demonstration of frequency dependency and accumulation of pulses to induce collective pacemaker activity similar to that observed *in vitro*.**

(A) Monolayers of optogenetically modified NRVMs show that the transition from the stationary state (STA) towards the oscillatory state (OSC) is dependent on the frequency of excitatory waves passing through an illuminated area under a constant number of 4 pulses. From left to right: no induction at stimulation periods of 150 and 1000 *ms*, induction at a stimulation period of 500 *ms*. **(B)** Monolayers of optogenetically modified NRVMs show that the transition from the stationary state towards the oscillatory state is dependent on the number of excitatory waves passing through an illuminated area under a constant frequency (1.666 *Hz* or a 600-*ms* period). From left to right: no induction at 2 pulses, induction at 4 pulses. **(C)** Three-dimensional parameter scan of all variables (light intensity, pacing period, and number of pulses) showing how dependency on period and on number of pulses relate to each other. Representative slices for fixed light intensities are displayed at the left. Vertical green lines show the natural pacemaker frequency the monolayer settles to after initiation of pacemaker activity. Letters in the middle panel correspond to traces in **A** and **B. (D)** Three-dimensional parameter diagram in which the size of the illuminated area was varied instead of the light intensity. Representative slices for fixed area edge lengths are displayed at the left. Vertical green lines show the natural pacemaker frequency the monolayer settles to after initiation of pacemaker activity.

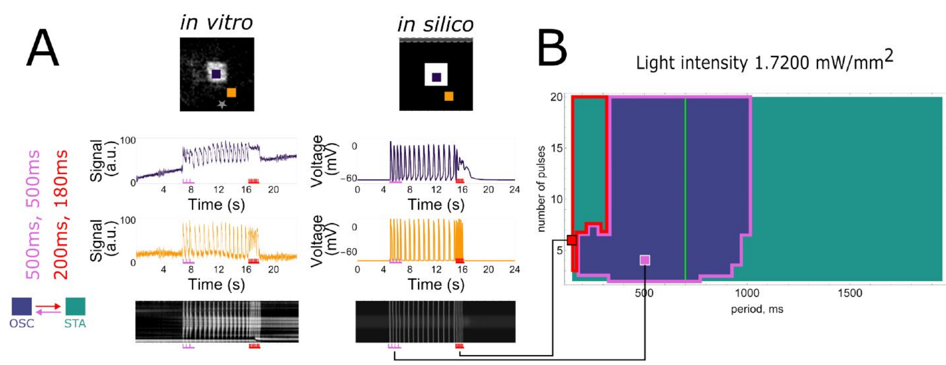


Figure 5.

Demonstration of frequency dependency to terminate collective pacemaker activity.

(A) Initiation and termination of collective pacemaker activity *in vitro* (left panel, 3 pulses of 500-ms period for initiation, 8 pulses of 200-ms period for termination) and *in silico* (right panel, 4 pulses of 500-ms period for initiation, 7 pulses of 180-ms period for termination). **(B)** Rightmost slice from **Figure 4C** showing *in silico* experiments at a fixed light intensity (1.72 mW/mm^2) and size of the illuminated area (67 pixels edge length) with indicated termination (red border) and initiation (magenta border) period ranges. Vertical green line shows the natural pacemaker frequency the monolayer settles to after initiation of pacemaker activity.

The collective nature of wave train-induced pacemaker activity

After exploring collective bi-stability in practice, we aimed to gain a deeper understanding of this phenomenon. Bi-stability is a wide-spread phenomenon and can take place at the single cell level (Guevara (2003)), which would be the simplest assumption for our monolayer system as well. When this would be the case, all cells would become oscillatory under illumination. We tested this hypothesis by making use of one of the four parameters we investigated earlier: the illuminated area. This parameter shows a transition from the stationary state towards the oscillatory state (**Figure 6A**) just like the light intensity parameter (**Figure 1B**) resulting in spontaneous oscillatory behaviour with waves emanating from the center of the illuminated area for a large area size and at a constant low light intensity (**Figure 4D**). These circumstances allowed us to compare the voltage traces between the center of the irradiated area and the bulk of the tissue, both *in vitro* and *in silico*. While the cells outside of the illuminated area showed oscillating membrane potentials those in the center of the illuminated region, i.e. those least affected by the coupling current of neighbouring cells, displayed a stable elevation of membrane potential. Hence, this implies that not all illuminated cells became oscillatory. We can therefore reject the simplest hypothesis of bi-stability at the single cell level, but have to speak of a collective phenomenon. It is for this reason that throughout the manuscript the observed effects are referred to as “collective” pacemaker activity.

Insight into pulse-dependent collective state transitions

Knowing that we are dealing with a collective phenomenon, we wanted to gain a deeper understanding in the observed pulse-dependent collective transitions from stationary towards oscillatory pacemaker activity. In order to achieve this goal, *in silico* analysis was performed (**Figure 7**) because it offers the possibility to visualise all parameters relevant for the system. In every monolayer, we selected a single point from the centre of the illuminated region of the tissue and looked at its dynamic behaviour in phase space.

As was shown in all figures from **Figure 2** to **Figure 6**, illuminated cells do not show dampening oscillations when reaching the stationary state after a transition from the oscillatory state. This behaviour is justified when looking at a projection of the phase space of the model where the voltage V and the gating variable x_r for the rapid delayed outward rectifier K^+ current were chosen for the x- and y-axis, respectively (**Figure 7A**). The stable fixed point (STA) is not located inside the stable limit cycle (OSC) (right panel), hence showing no Hopf bifurcation (left panel) with its associated damped oscillations. The two attractor regions (STA and OSC) were also reconstructed from experimental data and showed two distinct trajectories (**Appendix 2—figure 6**).

Since we are dealing with a high-dimensional phase space in the *in silico* model, the visualisation of trajectories in time and the corresponding dynamic behaviour is rather complex. However, from the two-dimensional projections onto the V and x_r variables, we can see what causes the frequency and pulse dependency (**Figure 2**, **Figure 3** and **Figure 4**) of ectopic pacemaker activity initiation, as visualised in **Figure 7C**. Four different pairs of stimulation period and number of pulses were selected, corresponding to different behaviour areas in the parameter space slice for a fixed illumination intensity (1.72 mW/mm^2) and size of the illuminated area (67 pixels edge length) (**Figure 7B**). The first point (α) comes from the region that shows no transition towards collective pacemaker activity for a low number of pulses (2) and intermediate stimulation periods of 800 ms. The phase space projection (**Figure 7C**) shows that the pulses cycle around the stable limit cycle, and return back to the stable fixed point, indicating that two pulses are not enough to move within the attractor zone of the stable limit cycle. When the number of pulses is increased to 4 (β), the attractor zone of the stable limit cycle is reached from the outside, and collective ectopic pacemaker activity is induced. After switching to a faster

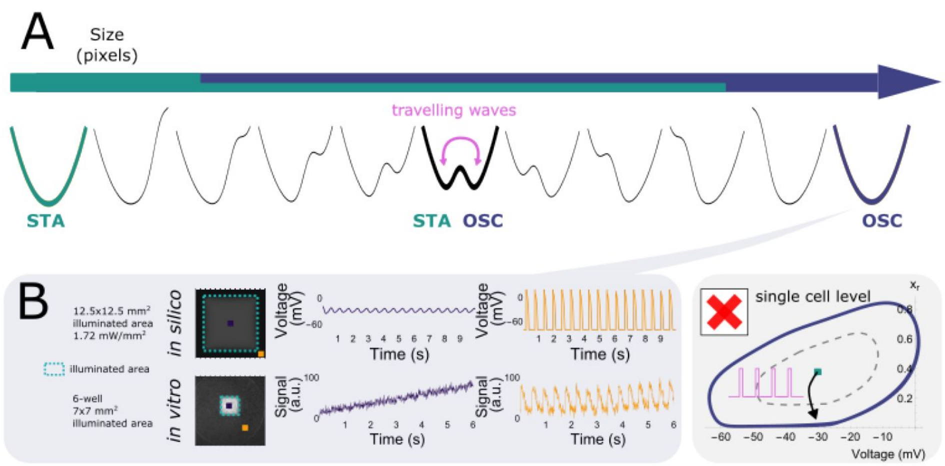


Figure 6.

Wave train-induced pacemaker activity is a multicellular collective phenomenon.

(A) Illuminated area (under constant light intensity) influences collective behaviour of excitable systems, transitioning between a stationary state for small illuminated areas and an oscillatory state for large illuminated areas. **(B)** In the oscillatory regime (large illuminated areas), illuminated cells in the center (purple traces) are depolarised (both *in silico* and *in vitro*), while oscillatory behaviour still takes place in the bulk of the tissue (orange traces). This discards the simplest model of bi-stable limit cycles at the single cell level.

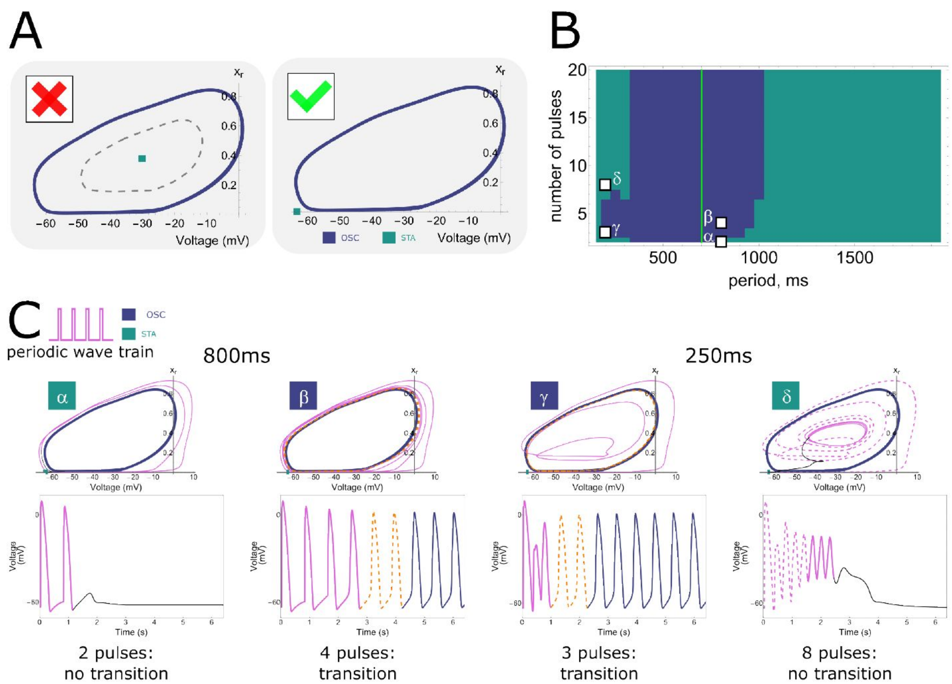


Figure 7.

Phase plane projections of pulse-dependent collective state transitions.

(A) Phase space trajectories (displayed in the Voltage – x_r plane) of the NRVM computational model show a limit cycle (OSC) that is not lying around a stable fixed point (STA). **(B)** Parameter space slice showing the relationship between stimulation period and number of pulses for a fixed illumination intensity (1.72 mW/mm^2) and size of the illuminated area (67 pixels edge length). Letters correspond to the graphs shown in **C. (C)** Phase space trajectories for different combinations of stimulus train period and number of pulses (α: 800 *ms* cycle length + 2 pulses, β: 800 *ms* cycle length + 4 pulses, γ: 250 *ms* cycle length + 3 pulses, δ: 250 *ms* cycle length + 8 pulses). α and δ do not result in a transition from the resting state to ectopic pacemaker activity, as under these circumstances the system moves towards the stationary stable fixed point from outside and inside the stable limit cycle, respectively. However, for β and γ, the stable limit cycle is approached from outside and inside, respectively, and ectopic pacemaker activity is induced.

stimulation period of 250 *ms* and sticking to just 3 pulses (γ), the phase space trajectories end up inside the stable limit cycle, but still within the attractor zone. Hence, also this time collective pacemaker activity was induced. If we now increase the number of pulses again, this time to 8 (δ), we are inside the stable limit cycle, but move again out of the region of attraction. Rather than producing oscillations, the system returns to the stationary state along dimensions other than those shown in **Figure 7C** (voltage and x_r), as evidenced by the phase space trajectory crossing itself. This return is mediated by the electrotonic current.

Thus, there is frequency selectivity due to the region of attraction of the stable limit cycle, where too low and too high frequencies result in stationary behaviour. Also pulse accumulation is shown, where more pulses move the system closer to the attractor region. However, this is not a Hopf bifurcation, because in that case the system would not return to the stationary state when the number of pulses exceeds a critical threshold.

Insight into the termination of ectopic pacemaker activity

Let us look once more at a cell inside the illuminated tissue undergoing a stimulus train with a short period of 180 *ms* from outside and extract all time-dependent variables. In response to a high number of stimuli (20 pulses), the time trace in the phase-space projection plot goes inside the limit cycle (**Figure 8A**) towards a quasi-steady-state, and returns back to the stationary state through a strangely curved line (black). This quasi-steady-state is hidden within the oscillatory trajectory and no pacemaker activity is initiated. When the number of pulses is increased from 20 to 40, the voltage trace shows exactly the same behaviour, indicating that the number of pulses has no influence on the mechanism that moves the tissue back to the stationary state (**Figure 8B**). However, repolarisation reserve does have an influence, prolonging the transition when it is reduced (**Appendix 2—figure 7**). This effect can be observed either by moving further from the boundary of the illuminated region where the electrotonic influence from the non-illuminated region is weaker, or by introducing ionic changes, such as a reduction in I_{Ks} and/or I_{to}. For example, because the electrotonic influence is weaker in the center of the illuminated region, the voltage there is not pulled down toward the resting membrane potential as quickly as in cells at the border of the illuminated zone.

To understand what is going on, we studied single cell dynamics under illumination. We already know from **Figure 6** that we are dealing with a purely collective phenomenon. Therefore, the single cell dynamics itself showed nothing peculiar, i.e. a standard current-voltage relationship (IV) curve with a single stable equilibrium (not shown). To add a multicellular component to our single cell model we introduced a current that replicates the effect of cell coupling and its associated electrotonic influence. This so-called bias current equals the average in time and space of the current inside the illuminated region using wave trains with a period of 180 *ms* (the current produced during the quasi-steady-state in **Figure 8B**). When this bias current is present, the IV curve of the single cell displays two stable equilibria and shows bi-stability (**Figure 8C-1**). The second stable equilibrium cannot be reached with a single pulse, but needs a stimulus train and can only be reached with short stimulation periods (180 *ms*) and intermediate stimulation amplitudes (-20 *mV*) (**Figure 8C-2**).

Building on this result, we next applied global stimuli with a period of 180 *ms* and amplitudes -58, -20, and 50 *mV* to a virtual cardiomyocyte monolayer with an illuminated square in the center (**Figure 8D**). These pulses were applied when the tissue was either in the oscillatory state (top two rows) or the stationary state (bottom two rows). The stimulus amplitude of -20 *mV* that was necessary to get the single cell with a bias current into the second stable equilibrium, also terminates oscillatory behaviour (**Figure 8D-2**) while the other two amplitudes of -58 and 50 *mV* (**Figure 8D-1,3**) don't alter the behaviour of the tissue. However, when starting from the stationary state, the behaviour is opposite and the intermediate amplitude of -20 *mV* fails to

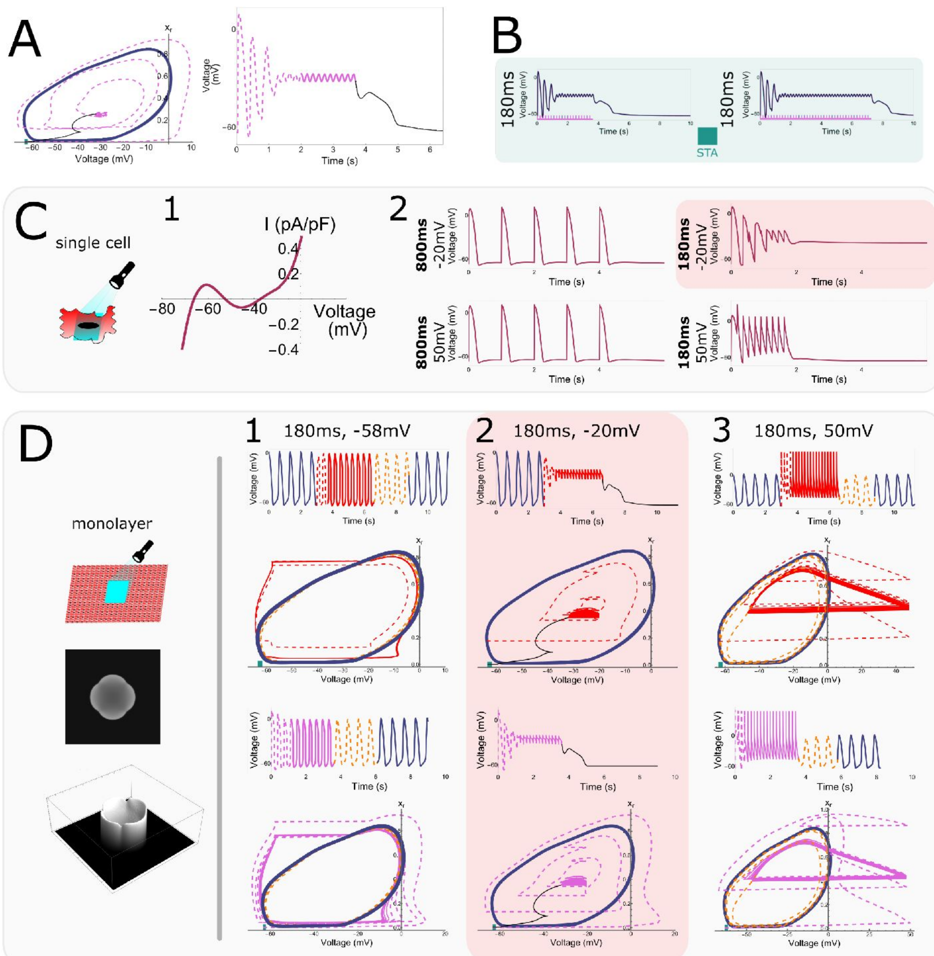


Figure 8.

Single cell "hidden" bi-stability and its relation to high frequency pacing.

(A) Phase space trajectory and corresponding voltage trace at short pacing period (180 *ms*). **(B)** Formation of the quasi-steady-state in the center of an irradiated region during a train of 20 and 40 high frequency (5.555 *Hz* or a 180-*ms* period) pulses. **(C)** Single cell bi-stability. (1) Single cell IV curve for an for illuminated cell (1.72 mW/mm^2) subject to a bias current. (2) Membrane voltage traces for stimulation periods of 1000 and 180 *ms* and stimulation amplitudes of -20 and 50 *mV*. A transition to the depolarised stable state occurred for short stimulation periods and low stimulation amplitudes. **(D)** Effect of spatially uniform voltage perturbations of different amplitudes (-58, -20, and 50 *mV*) and a period of 180 *ms* in cardiomyocyte monolayers. The top trace and phase space projection show the effect of perturbations when oscillation is ongoing, while the bottom ones show how perturbations affect the tissue is in the stationary resting state. At an amplitude of -20 *mV* (2) oscillations were terminated when ongoing and could not be initiated when the tissue was in the resting state. In the other two cases (1,3), the opposite happened.

induce collective oscillatory behaviour while the other two amplitudes do elicit this behaviour. We thus conclude that high stimulation frequencies and intermediate stimulus amplitudes are capable of terminating collective oscillatory behaviour by transitioning through a quasi-steady-state.

We call this kind of bi-stability capable of terminating collective oscillatory behaviour “*hidden bistability*” for two reasons: (1) Single cell bi-stability depends on electronic interactions. Hence it can be said that this bi-stability is “hidden” and only emerges in collective behaviour. (2) It’s possible to escape the region of attraction of the oscillatory trajectory through opposite paths.

It’s complex, but not as complex as it seems

While it may be believed that the observed phenomena and explanations are intrinsic to entangled interactions specific to the complexity of a biological sample (*in vitro*) and a realistic computational model (*in silico*), we show here that this is not the case. By reducing the model equations, we show that collective oscillatory behaviour is not limited to the specific case of NRVMs. We started from the initial NRVM equations and reduced them to contain only a single depolarising current (I_{Na}), two repolarising currents (I_{Kr} and the time-independent I_{K1}), and a spatially dependent optogenetic current (I_{ChR2}). Fast variables in these currents were assumed to keep their steady-state value, which caused them to be time-independent. The exact equations can be found in ***Appendix 1***.

Using this simplified system of three variables (the voltage V, the closing gate of the Na^+ channel h, and the opening gate of the K^+ channel x_r), we were able to reproduce all key features (frequency selectivity, pulse accumulation and bi-stability) of the more complex systems (**Figure 9**).

Just like in **Figure 7C**, the simplified model showed no initiation of collective pacemaker activity at a low number of 2 pulses and a low stimulation frequency of 0.444 Hz or a 2250-ms period or at a high number of 8 pulses and a high stimulation frequency of 1.666 Hz or a 600-ms period (**Figure 9A**). However, pacemaker activity was induced when applying 3 pulses and a low stimulation frequency or 2 pulses and a high stimulation frequency. At a low frequency, the phase space projection (again projected onto the V and x_r variable plane) shows that the limit cycle is approached from the outside, while at high frequencies it is approached from the inside. Altogether, these results show frequency selectivity and pulse accumulation in the simplified model (**Figure 9A**). To explain the findings in the rightmost panel of **Figure 9A**, we had a closer look at the single cell dynamics under high stimulation frequencies. As shown in **Figure 9B**, 4 pulses at a low frequency (1.000 Hz or a 1000-ms period) resulted in normal behaviour, but high frequency stimulation (2.000 Hz or a 500-ms period) pushed the cell towards a stably elevated membrane potential (**Figure 9B**). Looking at the influence of stimulus amplitude on the ability to get to this elevated membrane potential, it appears that after 7 pulses 3 of the 4 traces move towards the elevated membrane potential of -20 mV (**Figure 9C**). Panels B and C combined give us the same qualitative behaviour as seen in **Figure 8C**. The qualitative behaviours of **Figure 4A** and **Figure 8D** were also reproduced and can be found in **Appendix 2—figure 8**.

Discussion

In this study, *in vitro* observations were combined with *in silico* modelling to study and gain mechanistic insight into collective electrophysiological behaviour in NRVM monolayers.

- Firstly, we demonstrated the existence of transitions from stationary cardiac monolayers towards ones that show pacemaker activity coming from an illuminated region, which are induced by wave trains of external pulses (**Figure 1**) as a manifestation of **collective bi-stability**. This transition between the two states is dependent on four variables: the light intensity applied to the illuminated region, the size of the illuminated region, the number

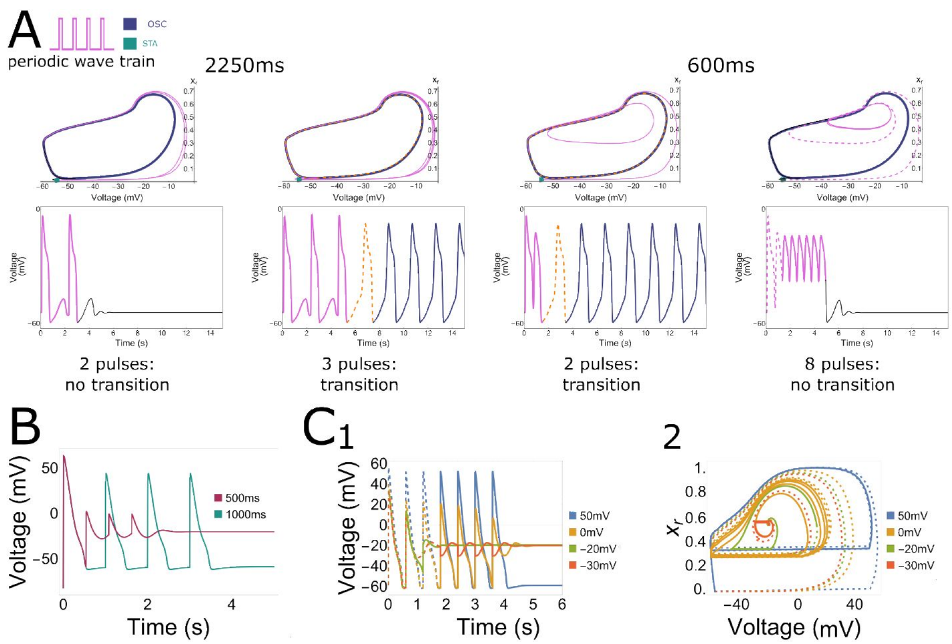


Figure 9.

Collective pacemaker activity in a simplified *in silico* NRVM model.

(A) Phase space trajectories of a point inside the illuminated region for different combinations of stimulus train period and number of pulses. A low stimulus frequency (0.444 *Hz* or a 2250-*ms* period) and a low number of 2 pulses, as well as a high stimulus frequency (1.666 *Hz* or a 600-*ms* period) with a high number of 8 pulses both result in no transition from the resting state to ectopic pacemaker activity, moving towards the stationary stable fixed point from outside and inside the stable limit cycle, respectively. However, when applying 3 pulses with a low frequency (0.444 *Hz* or a 2250-*ms* period) or 2 pulses with a high frequency (1.666 *Hz* or a 600-*ms* period), ectopic pacemaker activity is induced and the stable limit cycle is approached from outside and inside, respectively. **(B)** Under illumination, long period stimulation (1000 *ms*) shows a repolarised state after pacing, while short period stimulation shows a depolarised state (500 *ms*). **(C) 1** Evolution of the voltage variable as a result of high frequency voltage perturbations of high (blue trajectories) and low (green, orange and red) magnitudes. The dashed parts of graphs demonstrate the transient non-periodic parts of the trajectories, the solid lines show the periodic and stationary states. **2** Corresponding trajectories in the (V, x_r) phase plane.

of pulses of an incoming wave-train, and the frequency of the pulses of the incoming wave train (**Figure 2**, **Figure 3**, **Figure 5** and **Appendix 2—figure 4**, **Figure 5**). *In vitro* experiments were quantitatively reproduced *in silico*, thereby establishing a link between model and experiment (**Figure 4**, **Figure 5** and **Appendix 2—figure 4**, **Figure 5**)).

- Secondly, we studied the mechanisms underlying the initiation and termination of pacemaker activity. In the presence of a diffusion current (or bias current in the case of a single cell), the illuminated cells transition to a second equilibrium under high frequency pacing. Because this second equilibrium only appears in a coupled system (or in the presence of a bias current), we can also speak of a **hidden bi-stability**. Hence, hidden bi-stability (high vs. low stationary resting membrane) is shown to underlie the collective phenomena (stationary vs. oscillatory behaviour) (**Figure 6** **Figure 7** **Figure 8**, **Appendix 2—figure 6** and **Appendix 2—figure 7**).

Below, we compare these phenomena to different types of arrhythmic events and mechanisms of bi-stability in cardiac tissue, and extend the relevance of our mechanisms to spiking behaviour in neuronal and other biophysical systems.

Ectopic pacemaker activity in cardiology

The generation of ectopic beats is considered as one of the main mechanisms triggering and maintaining cardiac arrhythmias (Pogwizd et al. (1992)). In most cases, ectopic sources are believed to be a result of triggered activity, which can manifest itself as early afterdepolarizations (EADs) or delayed afterdepolarizations (DADs) (Qu et al. (2014)). Such ectopic sources are typically initiated by high frequency pacing resulting in a Ca^{2+} overload that triggers EADs and DADs (Lerman et al. (2024)). However, as we didn't observe Ca^{2+} overload, the phenomenon observed in this study resembles induced pacemaker activity instead. In contrast with ectopy, this pacemaker activity is initiated at low pacing frequencies. Therefore our findings might explain the phenomenon of bradycardia-induced triggered activity (Brachmann et al. (1983)). The initiation of oscillations in our monolayers is the result of the spatial dynamics of the system, which stands in contrast with the frequency dependency of EADs that can be explained on a single cell level (Tran et al. (2009); Qu et al. (2014)). Our optogenetic current is absent in normal cardiac tissue, but its associated depolarisation could be caused by other late activating inward currents, which are also essential in the induction of EADs or DADs (Song et al. (2008); Qu et al. (2014)): e.g. the late Na current ($I_{Na,L}$, Belardinelli et al. (2015)), reactivation of I_{CaL} (Tran et al. (2009)), or NCX activity (Song et al. (2015)). In addition, collective bi-stability might be observed in the presence of oxidative stress, ischemia or chronic heart failure (Belardinelli et al. (2015); Shryock et al. (2013); Undrovinas et al. (1999)), where there is depolarisation-induced automaticity (triggered activity) caused by a background current (e.g. $I_{Na,L}$).

The ectopic pacemaker activity in this study is a direct result of the combination of single cell depolarisation with spatial effects. Due to the additional inward current produced by the activation of CheRiff, a single uncoupled illuminated cell goes to a steady depolarised state (**Figure 6B**). However, in a multicellular environment there is a repolarising electrotonic coupling current working against this depolarising light-induced current. This repolarising current is a consequence of the interaction between depolarised and non-depolarised parts in the monolayer. By playing with the illumination intensity and the size of the illuminated region, we were able to balance these de- and re-polarising effects to create a bi-stable regime (STA vs. OSC). This also explains why there are no oscillations present in the center of a large illuminated region, since there the electrotonic current is lower than at its edges (**Figure 6C**). On the other hand, if the electrotonic current is too strong, it will prevent depolarisation by the optogenetic current and, as a consequence, the system wil stay in the stationary state. The balancing of currents also explains why the oscillatory boundary moves closer to the edge of the illuminated region at higher light intensities, since under those circumstances the depolarising current will overpower the spatial effects. However, when the illumination is very strong, it is possible to have oscillations that don't need to be induced (**Figure 1B**) and can even come from the corners of the illuminated region (**Appendix 2—figure 3**, Teplenin et al. (2018)). This last-mentioned phenomenon appears because the curvature of the boundary between illuminated and not-illuminated regions influences the electrotonic current.

Interplay between microscopic (single cell) and macroscopic (monolayer) behaviour

Collective bi-stability, as described in the previous section, is a bi-stability between two distinct behaviours (stationary STA and oscillatory OSC behaviour) that occur at the macroscopic level (monolayer). However, by looking more deeply at the dynamics that happen at the microscopic level (single cell), we can get more insight into the underlying mechanism(s). At the single cell level, the dynamic transition between a stationary monolayer and one showing induced pacemaker activity relies on two main processes:

1. For the initiation of pacemaker activity, pulse accumulation (at intermediate frequencies) is necessary to bring the system into the attracting region of the oscillatory state.
2. For the termination of pacemaker activity, an interaction takes place between "hidden" bistable dynamics and coupling currents at high frequencies.

Both these processes occur as a consequence of wave pulses outside of the illuminated region. At low frequencies, no oscillatory activity can be induced. At intermediate frequencies, it is possible to escape the attracting zone of the stationary state and cross the separatrix into the attracting zone of the oscillatory state. However, for even higher frequencies, the system gets pushed out of the attracting zone of the oscillatory state again and through hidden bi-stability it returns to the stationary state.

Our data on the termination mechanism of induced pacemaker activity (**Figure 8**) indicate that we are not dealing with classical STA-OSC bi-stability (Winfree (2001)). In systems displaying classical bi-stability between a limit cycle and the resting state, pacemaker annihilation can be achieved by applying sufficiently high frequency voltage perturbations. Through pulse accumulation it is then possible to exit the basin of attraction in the reverse direction from which it was entered. In contrast, our system enters (from outside the limit cycle) and leaves (towards the inside of the limit cycle) the attraction zone in opposite directions. This was concluded from the absence of a Hopf bifurcation in our system (**Figure 7**).

The seemingly fragile nature of hidden bi-stability might be a requirement for coexistence of excitable and bi-stable regimes. Otherwise, if the hills and dips of the N-shaped steady-state IV curve were large (**Figure 8C-1**), they would have similar magnitudes as the large currents of fast ion channels, preventing the subtle interaction between these strong ionic cell currents and the small repolarising coupling currents ($\approx 0.1\ \mathscr{pA}$).

Although the termination mechanism was shown to be amplitude-dependent (**Figure 8C-D**), the appropriate perturbation amplitudes are "automatically" selected by a high frequency train of travelling waves of excitation without any *a priori* knowledge about the state of the system. This wave train terminated ectopic triggers (**Figure 2**), which is in accordance with previous publications (Moak and Rosen (1984); Glikson et al. (2021)). These publications also showed that uniform high voltage shocks are unable to terminate ectopic sustained activity, similar to our findings (**Figure 8**).

Although its manifestations are clear in a monolayer system, hidden bi-stability might be challenging to be observed in single cell patch-clamp experiments. In **Figure 8C-1**, the height of dips and hills of the N-shaped steady-state IV curve under bias current are $\approx 0.2\mathscr{pA}/\mathscr{pF}$. While it is possible to perform patch-clamping on individual cells in a monolayer experiencing diffusion currents, measurements of small amplitudes can be hindered by the statistical and instrumental errors of voltage-clamp recordings. The voltage bi-stability we are looking for can, for example, easily be lost via a shift in current balance due to the non-zero resistance of the patch pipette in current clamp mode. This might explain why the phenomenon of hidden bi-stability has only rarely been reported, for example in an *in silico* study dedicated to myotony in muscle fibres (Cannon et al. (1993)).

Extrapolation and translation

All key aspects of the initiation and termination of induced pacemaker activity, i.e. the points itemized above, were also demonstrated in a simplified 3 variable reaction-diffusion model (**Figure 9**, **Appendix 2—figure 8**), implicating the more general nature of this observed phenomenon. A rigorous mathematical analysis can be accomplished in future work, but might be challenging due to the lack of efficient analytical tools to describe fully developed limit cycle oscillations in reaction-diffusion systems (Sherratt and Smith (2008)). Based on the reproduction of our results in such a simplified model, we showed that we don't need the full complexity of an ionic cardiomyocyte model, enabling extrapolation and translation of our results towards other scientific disciplines.

Frequency-dependent initiation and termination of oscillatory activity is not exclusive to cardiac systems, but is well known in a variety of non-linear biophysical systems. For example, the actomyosin cortex in the social amoeba *Dictyostelium discoideum* responds to short pulses of 3',5'-cyclic adenosine monophosphate (cAMP) by damped oscillations in average actin content, thereby allowing fine time-selective responses to either internal (pseudopodia initiation) or external (cAMP release) perturbations (Westendorf et al. (2013)). Moreover, in synthetic gene regulatory networks, subthreshold stimuli might be used to discriminate environmental stimuli (e.g. growth factor release, heat shock) (Guantes and Poyatos (2006)). Also neurons can differentiate external stimuli depending on their frequency. As mentioned in the introduction, the frequency-dependent initiation and termination of oscillatory activity is known as '*resonance*' in neuroscience. It is interesting to compare the resonance properties of our system with those actively studied in neurons.

Resonant neurons, also known as resonate-and-fire neurons or type II excitability neurons (Izhikevich (2000)), are governed by sub-threshold oscillations initiated due to proximity of the resting state to a Hopf bifurcation (Hutcheon and Yarom (2000); Roach et al. (2018)). They react to a stimulus train as follows: The first pulse initiates a damped sub-threshold oscillation of the membrane potential, while the effect of the second pulse depends on its timing. If timed correctly, it adds to the first pulse and either increases the amplitude of the sub-threshold oscillation or evokes repetitive spiking (transition to a limit cycle). Thus, the response and spiking of the resonator neuron depends on the frequency content of the input. These types of spiking neurons are usually associated with Bogdanov-Takens bifurcations (Izhikevich (2000)). However, another type of bifurcation, a saddle-node-loop (SNL) bifurcation, is also capable of generating spiking neurons and allows a bi-stability regime, in which the stable state lies outside the limit cycle similar to our case (**Figure 7**).

SNL bifurcations were previously identified in neurons (Hesse et al. (2017); Hesse and Schreiber (2019); Schleimer et al. (2021)), in a model of transcriptional regulation of the cell (Ciliberto et al. (2005)), and in the design of combined genetic switches (Perez-Carrasco et al. (2018)). Our result might represent the emergent analogue of a SNL bifurcation with resonant transitions on a neuronal/non-linear network level. SNL bifurcations have already been identified in the networks of coupled Kuramoto limit cycle oscillators both in a broad (Martens et al. (2009); ***Childs and Stro- gatz (2008)***) and in a neuroscience context (Buendía et al. (2021)). In these studies, SNL bifurcations were identified using dimensionality reduction techniques while ignoring the whole wealth of transient and finite-size phenomena. Moreover the Kuramoto model is a model without many physical details, describing a weak forcing regime on an idealized limit cycle oscillator, in which oscillations cannot be destroyed but only phase shifted. In our case, we provide a more realistic biophysically motivated model, which allows a strong forcing regime and destruction of oscillations due to the phenomenon of hidden bi-stability. This gives us reason to believe that idealized Kuramoto models could possess resonant properties if augmented/modified with sufficient physical/biophysical details. SNL bifurcations and hidden bi-stability could also be studied in a range of other unexplored biophysical situations including high frequency stimulation for the termination of neuronal oscillations during Parkinson disease (Touboul et al. (2020)), multi compartment cable computational models of the neuron (Schwemmer and Lewis (2012)), and bi-stability in neurons *in vitro* (***Le et al. (2006***)). Additional studies can be carried out in neuroscience by local optogenetic activation of depolarizing ion channels, both *in vivo*Lu et al. (2015) and *in silico* in non-local models Selvaraj et al. (2015); Heitmann et al. (2017).

Conclusion

Using both *in vitro* and *in silico* models, we explored a mechanism for frequency-dependent initiation and termination of oscillatory activity (i.e. induced pacemaker activity) in non-homogeneous cardiac tissue. This mechanism could be extended to more general excitable systems and is in neuroscience also known under the term resonance. Our collective “resonance” mechanism could be used to study or control collective states in cardiac but also other excitable systems.

Materials and Methods

Preparation of monolayers of NRVMs expressing the light-activatable cation channel CheRiff

All animal experiments were reviewed and approved by the Animal Experiments Committee of the Leiden University Medical Center and carried out in accordance with Directive 2010/63/EU of the European Union on the protection of animals used for scientific purposes. Monolayers of NRVMs were established as previously described Engels et al. (2015). Briefly, the hearts were excised from anaesthetised 2-day-old Wistar rats. The ventricles were cut into small pieces and dissociated in a solution containing 450 U/ml collagenase type I (Worthington, Lakewood, NJ) and 18,75 Kunitz/ml DNase I (Sigma-Aldrich, St. Louis, MO). The resulting cell suspension was enriched for cardiomyocytes by preplating for 120 minutes in a humidified incubator at 37°C and 5% CO_2 using Primaria culture dishes (Becton Dickinson, Breda, the Netherlands). Finally, the cells were seeded on round glass coverslips (d = 15 mm; Gerhard Menzel, Braunschweig, Germany) coated with bovine fibronectin (100 μg/ml; Sigma-Aldrich) to establish confluent monolayers. After incubation overnight in an atmosphere of humidified 95% air-5% CO_2 at 37°C, these monolayers were treated with Mitomycin-C (10 μg/ml; Sigma-Aldrich) for 2 hours to minimise proliferation of the non-cardiomyocytes. At day 4 of culture, the NRVM monolayers were incubated for 20-24 h with lentiviral vector particles encoding CheRiff (Hochbaum et al. (2014)), a light-gated depolarising ion channel of the channelrhodopsin family (Wang et al. (2015)), at a dose resulting in homogeneous transduction of nearly 100% of the cells (Feola et al. (2017); Ördög et al. (2023)). Next, the cultures were washed once with phosphate-buffer saline, given fresh culture medium and kept under culture conditions for 3-4 additional days.

Optical voltage mapping and patterned illumination of NRVM monolayers

After 8-10 days of culturing, NRVM monolayers were optically mapped using the voltage-sensitive dye di-4-ANBDQBS (52.5 μM final concentration; ITK diagnostics, Uithoorn, the Netherlands) as reported previously (Feola et al. (2017)). The mapping setup (**Appendix 2—figure 1**) was based on a 100 × 100 pixel CMOS Ultima-L camera (Scimedia, Costa Mesa, CA). The field of view was 16 × 16 mm resulting in a spatial resolution of 160 μm/pixel. For targeted illumination of monolayers the setup was optically conjugated to a digitally controlled micro-mirror device (DMD), the Polygon 400 (Mightex Systems, Toronto, ON), with a high power blue (470 nm) LED (BLS-LCS-0470-50-22-H, Mightex Systems). Before starting the actual experiments, all monolayers were mapped during 1-Hz electrical point stimulation to check baseline conditions. Electrical stimulation was performed by applying 10-ms-long rectangular electrical pulses with an amplitude of 8 V to a bipolar platinum electrode with a spacing of 1.5 mm between anode and cathode. Only cultures with an action potential duration (APD) at 80% repolarization (APD_{80}) below 350 *ms* and a conduction velocity (CV) above 18 *cm/s* were used for further experiments. Stimuli were applied in trains that varied in number and period. Whenever the outcome was reproducible three consecutive times in a single monolayer, it was counted as successful and included as one measurement in the total dataset. The constant light intensity was varied in the range of (0.03125-0.25 mW/mm^2) to search the critical value for the bi-stable regime. For experiments in which the light intensity was varied the rectangular-shaped area of illumination was in the range of 4 × 4 to 6 × 6 mm. The highest achievable irradiation intensity (0.3125 mW/mm^2) was used to perform size modulation experiments. The resulting electrical activity was recorded for 6-24 *s* at exposure times of 6 *ms* per frame.

Post-processing of optical mapping data

Data processing was performed using specialized BV Ana software (Scimedia), ImageJ (Schneider et al. (2012)) and custom-written scripts in Wolfram Mathematica (Wolfram Research, Hanborough, Oxfordshire, United Kingdom). APD and CV were calculated as described previously (Feola et al. (2017)). To prepare representative frames of wave propagation, optical mapping videos were fil-tered with a spatial averaging filter (3 × 3 stencil) and a derivative filter.

Attractor reconstruction

Multiple attractor reconstruction is based on Takens's embedding theorem (Takens (1981)) by plotting phase space trajectories in ($V(t)$, $V(t+\tau)$) coordinates. The optimal time delay τ was determined by calculating the minimum of the autocorrelation function (Clemson and Stefanovska (2014)). Linear drift removal was applied before plotting phase space trajectories.

Detailed electrophysiological model

In this study, the Majumder-Korhonen model (Majumder et al. (2016)) of NRVM monolayers was used, whose monodomain reaction-diffusion equation is formulated as follows:

$$\frac{\partial V}{\partial t} = \nabla \cdot (D \nabla V) - \frac{I_{ion} + I_{ChR2}(x,y)}{C_m}, \quad (1)$$

where I_{ion} is the sum of all ionic currents, $\mathscr{D}$ is the diffusion coefficient equal to 0.00095 cm^2/ms and $I_{ChR2}(x, y)$ is the spatially controlled optogenetic current. In comparison to the original model, I_{Na} was increased 1.3-fold and, I_{Ca_L} and I_{Kr} were both reduced 0.7-fold. The steady-state voltage de-pendency for the inactivation variable h of the fast Na^+ current was changed to $h_\infty = [1 + e^{\frac{(72+V)}{6.07}}]^{-1}$, where V is the transmembrane potential. The resulting CV was 20 cm/s. The optogenetic current I_{ChR2} was formulated based on the Boyle et al. (2013) I_{ChR2} model with the introduction of rectifying conduction properties and voltage-dependent state transitions (Williams et al. (2013)). The irradiation intensity varied in the range of 0.165–0.175 mW/mm^2.

The forward Euler method was used to integrate the equations with a time step Δt = 0.02ms and a centered finite-differencing scheme to discretise the Laplacian with a space step of Δx = 0.0625mm. The total computational domain size was 256 × 256 grid points; while the centrally irradiated square with the optogenetic current consisted of 80 × 80 grid points. To create stationary initial conditions, the model was integrated for 2 minutes before a train of stimuli was applied from the upper border of the domain.

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Additional information

Author contributions

Conceptualization: AST, AVP, DAP, TDC

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Software: AST, RM, NNK

Formal analysis: AST, TDC

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Resources: AAFdV, DAP

Writing—original draft: TDC

Writing—review & editing: AST, AAFdV, AVP, DAP, TDC

Visualization: AST, TDC

Supervision: DAP and TDC

Project administration: DAP

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Additional files

Supplemental Video 1

Appendix 1

Supplementary Methods

Simplified reaction-diffusion model

The simplified reaction-diffusion model consists of one voltage variable, two gating variables for the depolarising and hyperpolarising currents, and an inhomogeneity parameter $C(x, y)$:

$$\frac{\partial V}{\partial t} = D\Delta V - 6(1 + e^{-\frac{(45+V)}{6.5}})^{-3} j^2 (V - 62.84) \tag{2}$$

$$- (1 + e^{\frac{V+9}{22.4}})^{-1} x_r 0.011536(V + 86.8358) \tag{3}$$

$$- 0.6 I_{K1} - 0.24(10.6408 - 14.6408 e^{\frac{-V}{42.7671}}) C(x, y) \tag{4}$$

$$\frac{\partial j}{\partial t} = \frac{(j_\infty(V) - j)}{\tau_j(V)} \tag{5}$$

$$\frac{\partial x_r}{\partial t} = \frac{(x_{r_\infty}(V) - x_r)}{\tau_j(V)} \tag{6}$$

where D is the diffusion coefficient equal to 0.0003 cm^2/ms. The I_{K1} current formulation was copied from Majumder et al. (2016). The $C(x, y)$ was set to 0.1518115 inside the illuminated square and to 0 outside this. $j_\infty(V)$, $x_{r_\infty}(V)$, $\tau_j(V)$, $\tau_j(V)$ are described as:

$$j_\infty(V) = (1 + e^{\frac{(68+V)}{6.07}})^{-1} \tag{7}$$

$$x_{r_\infty}(V) = (1 + e^{\frac{-(21.5+V)}{7.5}})^{-1} \tag{8}$$

$$\tau_{x_r}(V) = \begin{cases} 2(\frac{0.00061(V+38.9)}{e^{0.145(V+38.9)}-1} + \frac{0.00138(V+14.2)}{1-e^{-0.123(V+14.2)}})^{-1}, V > -57 \\ 4(\frac{0.00061(V+38.9)}{e^{0.145(V+38.9)}-1} + \frac{0.00138(V+14.2)}{1-e^{-0.123(V+14.2)}})^{-1}, V \le -57 \end{cases} \tag{9}$$

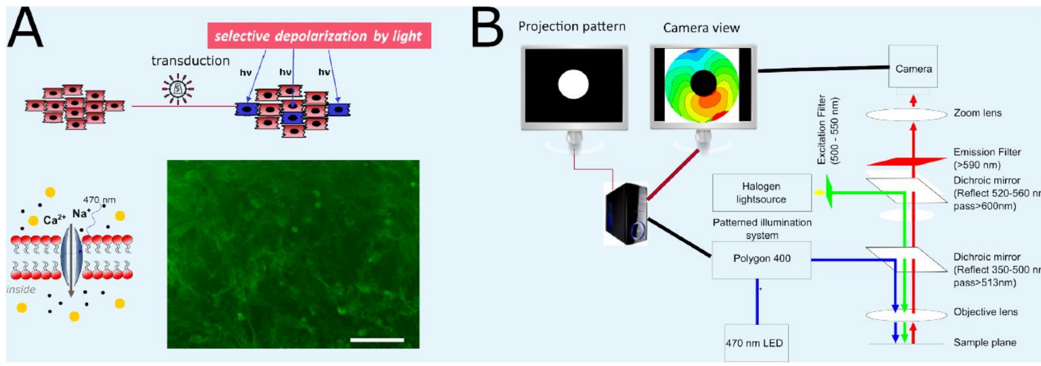
$$\tau_j(V) = \begin{cases} 11.63(1 + e^{-0.1(V+32)}) e^{2.535*10^{-7} V}, V \ge -40 \\ 3.49(\frac{-127140(V+37.78)}{1+e^{0.311(V+79.23)}} e^{0.2444V} - \frac{3.474}{10^5 e^{0.04391*V}} + \frac{0.1212 e^{-0.01052*V}}{1+e^{0.1378(V+40.14)}})^{-1}, V < -40 \end{cases} \tag{10}$$

(11)

The forward Euler method was used to integrate the equations with a time step $\Delta t = 0.01 ms$ and a centered finite-differencing scheme to discretise the Laplacian with a space step of $\Delta x = 0.0625 mm$. The total computational domain size was 256 × 256 grid points while the square with the “background” current consisted of 80 × 80 grid points. To create stationary initial conditions, the model was integrated for 2 minutes before a train of stimuli was applied from the upper border of the domain.

Appendix 2

Supplementary Figures



Appendix 2—figure 1.

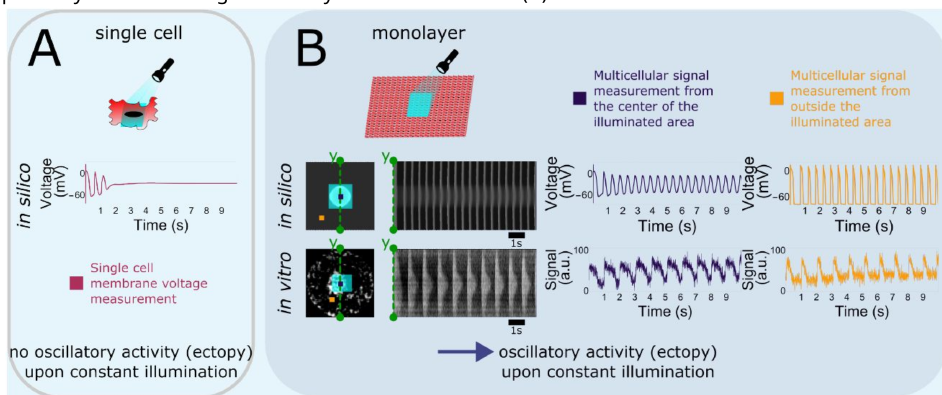
Experimental set-up.

(A) Diagrams of the *in vitro* model of NRVMs expressing a light-gated ion channel (top) and of the depolarising light-gated ion channel CheRiff (bottom) together with a representative example of the cell surface fluorescence produced by the enhanced yellow fluorescent protein tag attached to the C terminus of the light-gated ion channel in a CheRiff-expressing NRVM monolayer. **(B)** Optical voltage mapping setup including mapping computer, patterned illuminator (Polygon 400), and optical mapping camera.

Appendix 2—figure 2.

Optogenetic depolarising tools and ectopy.

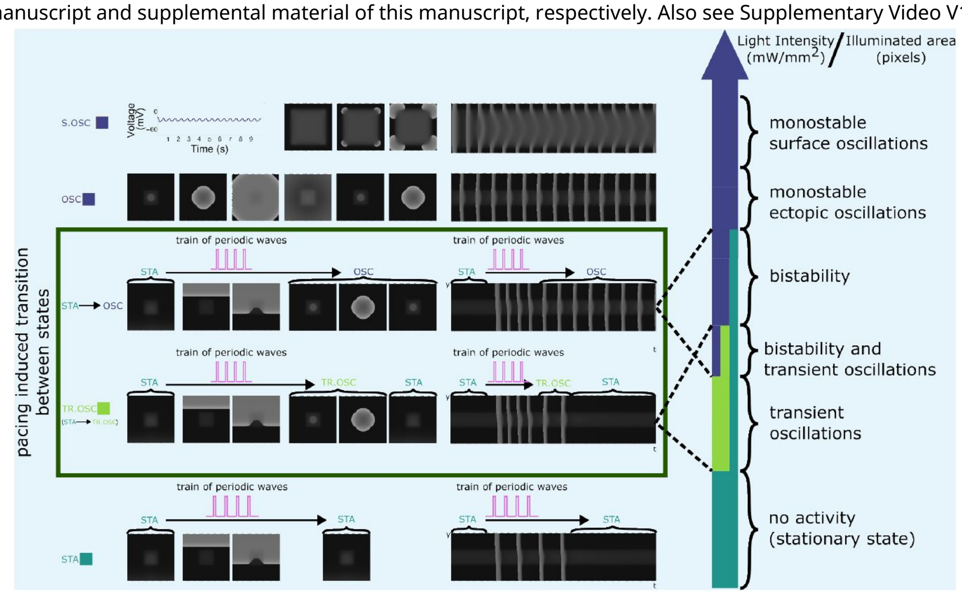
(A) Single cell computational data for application of an optogenetic depolarising tool under constant light illumination, showing a steady depolarised state and lack of sustained oscillations in transmembrane voltage. **(B)** Computational and experimental data showing emergence of ectopic activity in a coupled multicellular system. Sustained ectopic activity emanates periodically from the inside (center) of the illuminated area (light blue box) in both the computational and the experimental model. The sampling positions inside and outside of the illuminated area are indicated by purple and orange squares, respectively. The *in silico* light intensity was the same as in (A).

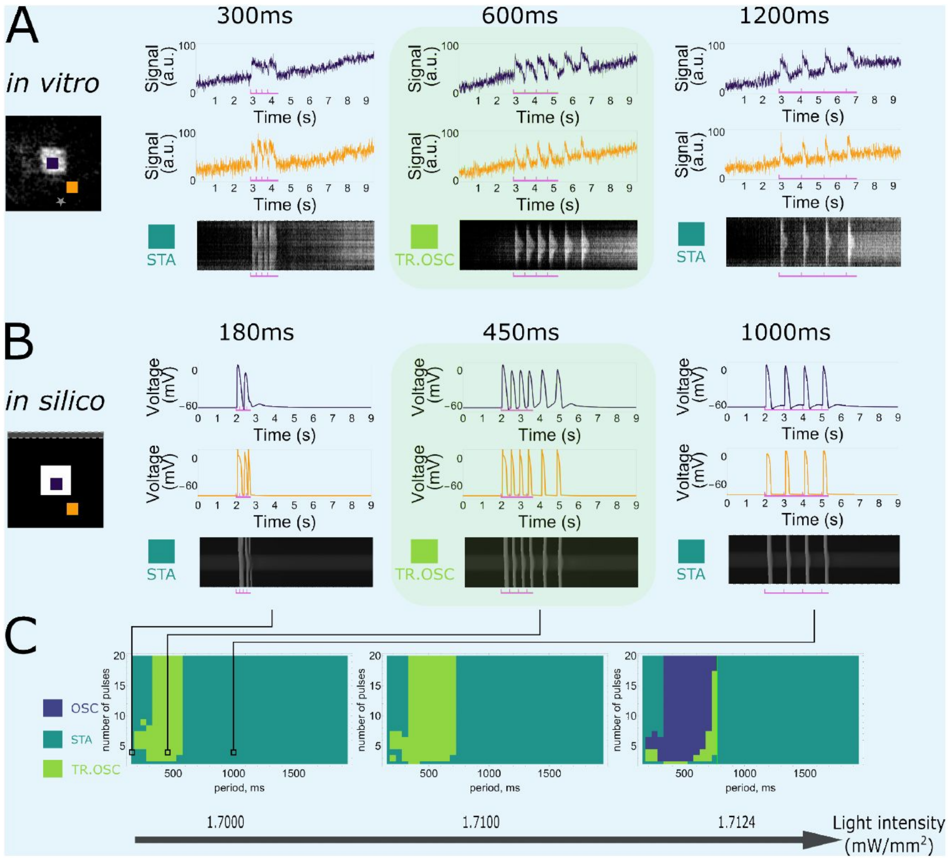


Appendix 2—figure 3.

A range of collective phenomena are observed in a multicellular *in silico* NRVM model.

Exposure of a squared region in the center of the NRVM monolayer to light of different intensities generates of a photocurrent of different strength by the light-gated depolarising ion channel. Application of a periodic wave train to these NRVM monolayers results in 5 different outcomes, which are each visualised by representative snapshots and a linear time-extract. Going from a high (top) to low (bottom) light intensity these outcomes are: (1) monostable surface oscillations, S.OSC, (2) monostable ectopic oscillations, OSC, (3) bi-stability, (4) transient oscillations, TR.OSC, and (5) stationary state with no activity, STA. pThe bi-stability region (STA->OSC) and the transient oscillations (STA->TR.OSC) (green box) are described in the main manuscript and supplemental material of this manuscript, respectively. Also see Supplementary Video V1.

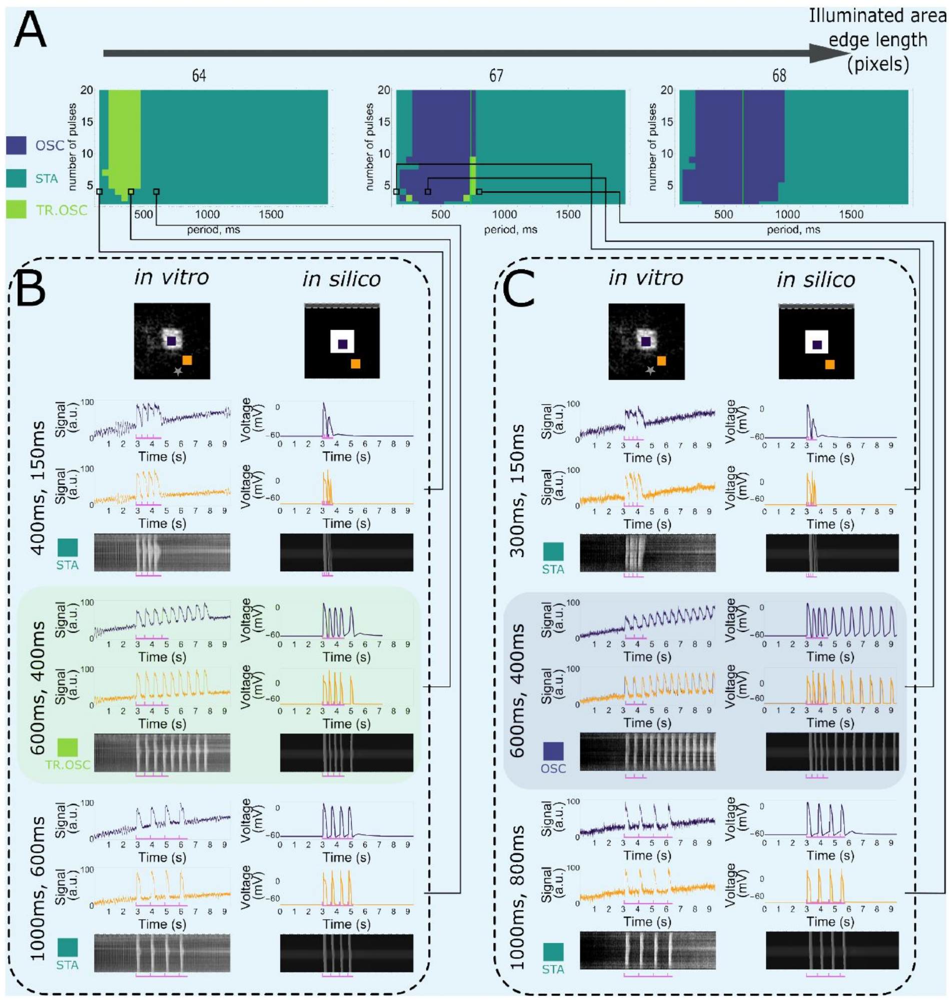




Appendix 2—figure 4.

Frequency dependency of transient pacemaker activity induction.

(A) *In vitro* monolayers of optogenetically modified NRVMs show that the transition from the stationary state (STA) towards the transient oscillatory state (TR.OSC) is dependent on the frequency of excitatory waves passing through an illuminated area under a constant number of 4 pulses. Four pulses at a high frequency (3.333 *Hz* or a 300-*ms* period) or low frequency (0.833 *Hz* or a 1200-*ms* period) do not induce the transition in contrast to 4 pulses with an intermediate period of 600 *ms*. **(B)** Qualitative correspondence of this behaviour is shown in a computational NRVM model for low (1.000 *Hz* or a 1000-*ms* period), intermediate (2.222 *Hz* or a 450-*ms* period), and high (6.666 *Hz* or a 150-*ms* period) stimulation frequencies. **(C)** Slices of the full parameter plot for a constant size of the illuminated central region of the optogenetically modified NRVM monolayer (80 × 80 pixels) with indications of the selected frequency visualisations.



Appendix 2—figure 5.

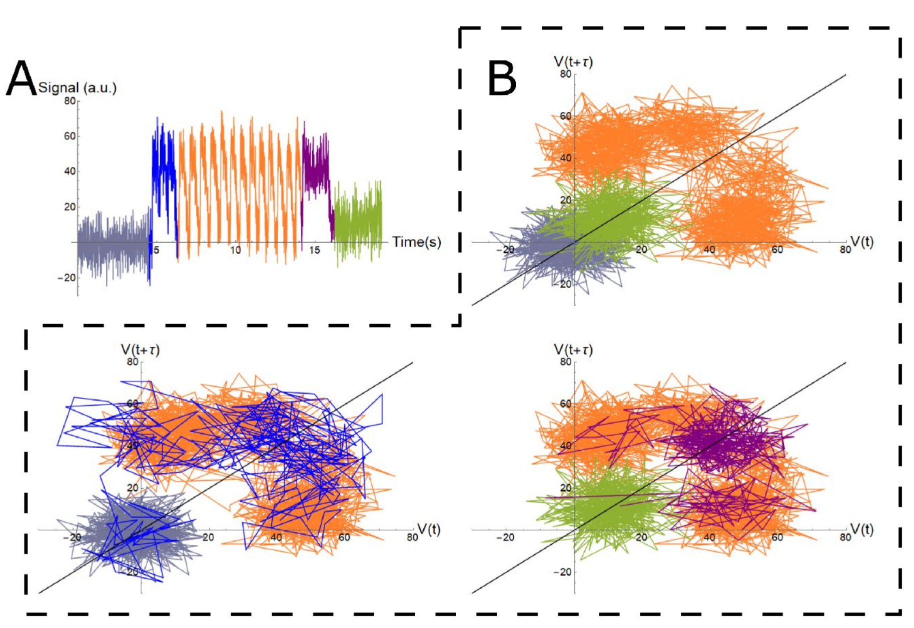
Influence of the size of the illuminated central square on transient pacemaker activity induction.

(A) Slices of the full parameter plot for constant illumination (1.72 mW/mm^2) with indications of selected frequency visualisations. **(B)** Transient oscillations can be induced with a small-sized illumination region (64 × 64 pixels) for intermediate frequencies. **(C)** The oscillatory state can be induced with a large-sized illumination region (67 × 67 pixels) for intermediate frequencies.

Appendix 2—figure 6.

Trajectory reconstruction using Takens time delay embedding.

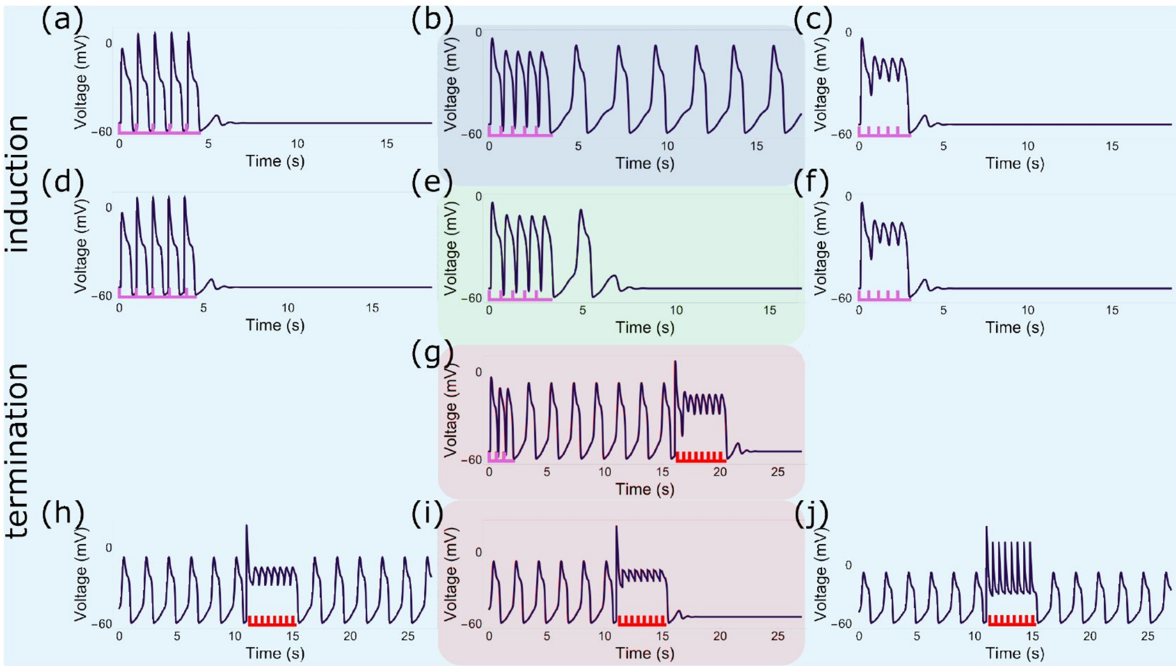
(A) The optical voltage signal shows induced initiation and termination of pacemaker activity. The initial stationary state is shown in grey, the initiating stimulating pulses in blue, periodic oscillations in orange, the terminating high frequency stimulation in purple, and the stationary state after termination of oscillatory pacemaker activity in light green. **(B)** Trajectories in phase space in ($V(t)$, $V(t+\tau)$) coordinates showing multiple attractors (grey/green [STA]) and orange [OSC]) and the transition between them (blue/purple).



Appendix 2—figure 7.

The effect of ionic changes on the termination of pacemaker activity

The mechanism that moves the oscillating illuminated tissue back to the stationary state after high frequency pacing (20 and 10 pulses at 5.555 *Hz* or 180-*ms* periods) is dependent on the ionic properties of the tissue, i.e. lower repolarisation reserves (20% I_{Ks} + 50% I_{to}) are associated with longer transition times.



Appendix 2—figure 8.

Qualitative similarities between the simplified and complex reaction-diffusion model.

(a-c) Five pulses of low (1.064 *Hz* or a 940-*ms* period) and high (1.852 *Hz* or a 540-*ms* period) frequency at a light intensity of 0.1518121 mW/mm^2 fail to induce pacemaker activity in the simplified *in silico* NRVM model, while intermediate frequencies (1.515 *Hz* or a 660-*ms* period) do. (d-f) Under lower illumination intensity (0.15181133 mW/mm^2), five pulses of low (1.064 *Hz* or a 940-*ms* period) and high (1.852 *Hz* or a 540-*ms* period) frequency fail to induce pacemaker activity, while intermediate frequencies (1.515 *Hz* or a 660-*ms* period) induce transient oscillations. (g) Termination of pacemaker activity by high frequency travelling waves of excitation (initiate: 3 pulses at a 660-*ms* period, terminate: 8 pulses at a 540-*ms* period, light intensity 0.1518121 mW/mm^2). (h-j) Failure to terminate pacemaker activity by spatially uniform voltage perturbations of high frequency (1.852 *Hz* or a 540-*ms* period, light intensity 0.1518121 mW/mm^2) and low (-30 *mV*) or high (50 *mV*) magnitude, while termination is successful at an intermediate magnitude of -20 *mV*.

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Reviewer #1 (Public review):

Summary:

The study by Teplenin and coworkers assesses the combined effects of localized depolarization and excitatory electrical stimulation in myocardial monolayers. They study the electrophysiological behaviour of cultured neonatal rat ventricular cardiomyocytes expressing the light-gated cation channel Cheriff, allowing them to induce local depolarization of varying area and amplitude, the latter titrated by the applied light intensity. In addition, they used computational modeling to screen for critical parameters determining state transitions, and for dissecting the underlying mechanisms. Two stable states, thus bistability, could be induced upon local depolarization and electrical stimulation, one state characterized by a constant membrane voltage and a second spontaneously firing, thus oscillatory state. The resulting 'state' of the monolayer was dependent on the duration and frequency of electrical stimuli, as well as the size of the illuminated area and the applied light intensity determining the degree of depolarization as well as the steepness of the local voltage gradient. In addition to the induction of oscillatory behaviour, they also tested frequency-dependent termination of induced oscillations.

Strengths:

The data from optogenetic experiments and computational modelling provide quantitative insights into the parameter space determining the induction of spontaneous excitation in the monolayer. The most important findings can also be reproduced using a strongly reduced computational model, suggesting that the observed phenomena might be more generally applicable.

Weaknesses:

While the study is thoroughly performed and provides interesting mechanistic insights into scenarios of ventricular arrhythmogenesis in the presence of localized depolarized tissue areas, the translational perspective of the study remains relatively vague. In addition, the chosen theoretical approach and the way the data is presented might make it difficult for the wider community of cardiac researchers to understand the significance of the study.

Comments on Revision:

The provided revisions address some of the raised concerns, but they do not change my general assessment of the paper, including its strengths and weaknesses.

https://doi.org/10.7554/eLife.107072.2.sa2

Reviewer #2 (Public review):

In the presented manuscript, Teplenin and colleagues use both electrical pacing and optogenetic stimulation to create a reproducible, controllable source of ectopy in cardiomyocyte monolayers. To accomplish this, they use a careful calibration of electrical pacing characteristics (i.e., frequency, number of pulses) and illumination characteristics (i.e., light intensity, surface area) to show that there exists a "sweet spot" where oscillatory excitations can emerge proximal to the optogenetically depolarized region following electrical pacing cessation, akin to pacemaker cells. Furthermore, the authors demonstrate that a high-frequency electrical wave-train can be used to terminate these oscillatory excitations. The authors observed this oscillatory phenomenon both in vitro (using neonatal rat ventricular cardiomyocyte monolayers) and in silico (using a computational action potential model of the same cell type). These are surprising findings and provide a novel approach for studying triggered activity in cardiac tissue.

The study is extremely thorough and one of the more memorable and grounded applications of cardiac optogenetics in the past decade. One of the benefits of the authors' "two-prong" approach of experimental preps and computational models is that they could probe the number of potential variable combinations much deeper than through in vitro experiments alone. The strong similarities between the real-life and computational findings suggest that these oscillatory excitations are consistent, reproducible, and controllable.

Triggered activity, which can lead to ventricular arrhythmias and cardiac sudden death, has been largely contributed to sub-cellular phenomena, such as early or delayed afterdepolarizations, and thus to date has largely been studied in isolated single cardiomyocytes. However, these findings have been difficult to translate to tissue- and organ-scale experiments, as well-coupled cardiac tissue has notably different electrical properties. This underscores the significance of the study's methodological advances: use of a constant depolarizing current in a subset of (illuminated) cells to reliably result in triggered activity could facilitate the more consistent evaluation of triggered activity at various scales. An experimental prep that is both repeatable and controllable (i.e., both initiated and terminated through the same means) is a boon for further inquiry.

The authors also substantially explored phase space and single cell analyses to document how this "hidden" bi-stable phenomenon can be uncovered during emergent collective tissue behavior. Calibration and testing of different aspects (e.g.: light intensity, illuminated surface area, electrical pulse frequency, electrical pulse count) and other deeper analyses, as illustrated in Figures S3-S8 and Video S1, are significant and commendable.

Given the study is computational, it is surprising that the authors did not replicate their findings using well-validated adult ventricular cardiomyocyte action potential models, such ten Tusscher 2006 or O'Hara 2011. This may have felt out-of-scope, given the nice alignment of rat cardiomyocyte data between in vitro and in silico experiments. However, it would have been helpful peace-of-mind validation, given the significant ionic current differences between neonatal rat and adult ventricular tissue. It is not fully clear whether the pulse trains could have resulted in the same bi-stable oscillatory behavior, given the longer APD of humans relative to rats. The observed phenomenon certainly would be frequency-dependent and would have required tedious calibration for a new cell type, albeit partially mitigated by the relative ease of in silico experiments.

There are likely also mechanistic differences between this optogenetically-tied oscillatory behavior and triggered activity observed in other studies. This is because the constant light-elicited depolarizing current is disrupting the typical resting cardiomyocyte state, thereby altering the balance between depolarizing ionic currents (such as Na+ and Ca2+) and repolarizing ionic currents (such as K+ and Ca2+). The oscillatory excitations appear to later emerge at the border of the illuminated region and non-stimulated surrounding tissue, which is likely an area of high source-sink mismatch. The authors appear to acknowledge differences in this oscillatory behavior and previous sub-cellular triggered activity research in their discussion of ectopic pacemaker activity, which are canonically observed in genetic, pharmacologic, or pathological ionic conditions. Regardless, it is exciting to see new ground being broken in this difficult-to-characterize experimental space, even if the method illustrated here may not necessarily be broadly applicable.

Comments on revisions:

I have read the authors' rebuttal to our earlier comments and do not have any further questions or comments. Thank you for implementing the minor improvements to Figure visualizations and for creating Video S1 to accompany the article.

https://doi.org/10.7554/eLife.107072.2.sa1

Author response:

The following is the authors’ response to the original reviews

> ***Public Reviews:***
>
> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *The study by Teplenin and coworkers assesses the combined effects of localized depolarization and excitatory electrical stimulation in myocardial monolayers. They study the electrophysiological behaviour of cultured neonatal rat ventricular cardiomyocytes expressing the light-gated cation channel Cheriff, allowing them to induce local depolarization of varying area and amplitude, the latter titrated by the applied light intensity. In addition, they used computational modeling to screen for critical parameters determining state transitions and to dissect the underlying mechanisms. Two stable states, thus bistability, could be induced upon local depolarization and electrical stimulation, one state characterized by a constant membrane voltage and a second, spontaneously firing, thus oscillatory state. The resulting 'state' of the monolayer was dependent on the duration and frequency of electrical stimuli, as well as the size of the illuminated area and the applied light intensity, determining the degree of depolarization as well as the steepness of the local voltage gradient. In addition to the induction of oscillatory behaviour, they also tested frequency-dependent termination of induced oscillations.*
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> *Strengths:*
>
> *The data from optogenetic experiments and computational modelling provide quantitative insights into the parameter space determining the induction of spontaneous*

excitation in the monolayer. The most important findings can also be reproduced using a strongly reduced computational model, suggesting that the observed phenomena might be more generally applicable.

Weaknesses:

While the study is thoroughly performed and provides interesting mechanistic insights into scenarios of ventricular arrhythmogenesis in the presence of localized depolarized tissue areas, the translational perspective of the study remains relatively vague. In addition, the chosen theoretical approach and the way the data are presented might make it difficult for the wider community of cardiac researchers to understand the significance of the study.

Reviewer #2 (Public review):

In the presented manuscript, Teplenin and colleagues use both electrical pacing and optogenetic stimulation to create a reproducible, controllable source of ectopy in cardiomyocyte monolayers. To accomplish this, they use a careful calibration of electrical pacing characteristics (i.e., frequency, number of pulses) and illumination characteristics (i.e., light intensity, surface area) to show that there exists a "sweet spot" where oscillatory excitations can emerge proximal to the optogenetically depolarized region following electrical pacing cessation, akin to pacemaker cells. Furthermore, the authors demonstrate that a high-frequency electrical wave-train can be used to terminate these oscillatory excitations. The authors observed this oscillatory phenomenon both in vitro (using neonatal rat ventricular cardiomyocyte monolayers) and in silico (using a computational action potential model of the same cell type). These are surprising findings and provide a novel approach for studying triggered activity in cardiac tissue.

The study is extremely thorough and one of the more memorable and grounded applications of cardiac optogenetics in the past decade. One of the benefits of the authors' "two-prong" approach of experimental preps and computational models is that they could probe the number of potential variable combinations much deeper than through in vitro experiments alone. The strong similarities between the real-life and computational findings suggest that these oscillatory excitations are consistent, reproducible, and controllable.

Triggered activity, which can lead to ventricular arrhythmias and cardiac sudden death, has been largely attributed to sub-cellular phenomena, such as early or delayed afterdepolarizations, and thus to date has largely been studied in isolated single cardiomyocytes. However, these findings have been difficult to translate to tissue and organ-scale experiments, as well-coupled cardiac tissue has notably different electrical properties. This underscores the significance of the study's methodological advances: the use of a constant depolarizing current in a subset of (illuminated) cells to reliably result in triggered activity could facilitate the more consistent evaluation of triggered activity at various scales. An experimental prep that is both repeatable and controllable (i.e., both initiated and terminated through the same means).

The authors also substantially explored phase space and single-cell analyses to document how this "hidden" bi-stable phenomenon can be uncovered during emergent collective tissue behavior. Calibration and testing of different aspects (e.g., light intensity, illuminated surface area, electrical pulse frequency, electrical pulse count) and other deeper analyses, as illustrated in Appendix 2, Figures 3-8, are significant and commendable.

Given that the study is computational, it is surprising that the authors did not replicate their findings using well-validated adult ventricular cardiomyocyte action potential models, such as ten Tusscher 2006 or O'Hara 2011. This may have felt out of scope, given

> *the nice alignment of rat cardiomyocyte data between in vitro and in silico experiments. However, it would have been helpful peace-of-mind validation, given the significant ionic current differences between neonatal rat and adult ventricular tissue. It is not fully clear whether the pulse trains could have resulted in the same bi-stable oscillatory behavior, given the longer APD of humans relative to rats. The observed phenomenon certainly would be frequency-dependent and would have required tedious calibration for a new cell type, albeit partially mitigated by the relative ease of in silico experiments.*
>
> *For all its strengths, there are likely significant mechanistic differences between this optogenetically tied oscillatory behavior and triggered activity observed in other studies. This is because the constant light-elicited depolarizing current is disrupting the typical resting cardiomyocyte state, thereby altering the balance between depolarizing ionic currents (such as Na+ and Ca2+) and repolarizing ionic currents (such as K+ and Ca2+). The oscillatory excitations appear to later emerge at the border of the illuminated region and non-stimulated surrounding tissue, which is likely an area of high source-sink mismatch. The authors appear to acknowledge differences in this oscillatory behavior and previous sub-cellular triggered activity research in their discussion of ectopic pacemaker activity, which is canonically expected more so from genetic or pathological conditions. Regardless, it is exciting to see new ground being broken in this difficult-to-characterize experimental space, even if the method illustrated here may not necessarily be broadly applicable.*

We thank the reviewers for their thoughtful and constructive feedback, as well as for recognizing the conceptual and technical strengths of our work. We are especially pleased that our integrated use of optogenetics, electrical pacing, and computational modelling was seen as a rigorous and innovative approach to investigating spontaneous excitability in cardiac tissue.

At the core of our study was the decision to focus exclusively on neonatal rat ventricular cardiomyocytes. This ensured a tightly controlled and consistent environment across experimental and computational settings, allowing for direct comparison and deeper mechanistic insight. While extending our findings to adult or human cardiomyocytes would enhance translational relevance, such efforts are complicated by the distinct ionic properties and action potential dynamics of these cells, as also noted by Reviewer #2. For this foundational study, we chose to prioritize depth and clarity over breadth.

Our computational domain was designed to faithfully reflect the experimental system. The strong agreement between both domains is encouraging and supports the robustness of our framework. Although some degree of theoretical abstraction was necessary (thereby sometimes making it a bit harder to read), it reflects the intrinsic complexity of the collective behaviours we aimed to capture such as emergent bi-stability. To make these ideas more accessible, we included simplified illustrations, a reduced model, and extensive supplementary material.

A key insight from our work is the emergence of oscillatory behaviour through interaction of illuminated and non-illuminated regions. Rather than replicating classical sub-cellular triggered activity, this behaviour arises from systems-level dynamics shaped by the imposed depolarizing current and surrounding electrotonic environment. By tuning illumination and local pacing parameters, we could reproducibly induce and suppress these oscillations, thereby providing a controllable platform to study ectopy as a manifestation of spatial heterogeneity and collective dynamics.

Altogether, our aim was to build a clear and versatile model system for investigating how spatial structure and pacing influence the conditions under which bistability becomes apparent in cardiac tissue. We believe this platform lays strong groundwork for future extensions into more physiologically and clinically relevant contexts.

In revising the manuscript, we carefully addressed all points raised by the reviewers. We have also responded to each of their specific comments in detail, which are provided below.

> ***Recommendations for the Authors:***
>
> ***Reviewer #1 (Recommendations for the authors):***
>
> *Please find my specific comments and suggestions below:*
>
> *(1) Line 64: When first introduced, the concept of 'emergent bi-stability' may not be clear to the reader.*

We concur that the full breadth of the concept of emergent bi-stability may not be immediately clear upon first mention. Nonetheless, its components have been introduced separately: “emergent” was linked to multicellular behaviour in line 63, while “bi-stability” was described in detail in lines 39–56. We therefore believe that readers could form an intuitive understanding of the combined term, which will be further clarified as the manuscript develops. To further ease comprehension of the reader, we have added the following clarification to line 64:

“Within this dynamic system of cardiomyocytes, we investigated emergent bi-stability (a concept that will be explained more thoroughly later on) in cell monolayers under the influence of spatial depolarization patterns.”

> *(2) Lines 67-80: While the introduction until line 66 is extremely well written, the introduction of both cardiac arrhythmia and cardiac optogenetics could be improved. It is especially surprising that miniSOG is first mentioned as a tool for optogenetic depolarisation of cardiomyocytes, as the authors would probably agree that Channelrhodopsins are by far the most commonly applied tools for optogenetic depolarisation (please also refer to the literature by others in this respect). In addition, miniSOG has side effects other than depolarisation, and thus cannot be the tool of choice when not directly studying the effects of oxidative stress or damage.*

The reviewer is absolutely correct in noting that channelrhodopsins are the most commonly applied tools for optogenetic depolarisation. We introduced miniSOG primarily for historical context: the effects of specific depolarization patterns on collective pacemaker activity were first observed with this tool (Teplenin et al., 2018). In that paper, we also reported ultralong action potentials, occurring as a side effect of cumulative miniSOG-induced ROS damage. In the following paragraph (starting at line 81), we emphasize that membrane potential can be controlled much better using channelrhodopsins, which is why we employed them in the present study.

> *(3) Line 78: I appreciate the concept of 'high curvature', but please always state which parameter(s) you are referring to (membrane voltage in space/time, etc?).*

We corrected our statement to include the specification of space curvature of the depolarised region:

“In such a system, it was previously observed that spatiotemporal illumination can give rise to collective behaviour and ectopic waves (Teplenin et al. (2018)) originating from illuminated/depolarised regions (with high spatial curvature).”

> *(4) Line 79: 'bi-stable state' - not yet properly introduced in this context.*

The bi-stability mentioned here refers back to single cell bistability introduced in Teplenin et al. (2018), which we cited again for clarity.

"These waves resulted from the interplay between the diffusion current and the single cell bi-stable state (Teplenin et al. (2018)) that was induced in the illuminated region."

> *(5) Line 84-85: 'these ion channels allow the cells to respond' - please describe the channel used; and please correct: the channels respond to light, not the cells. Re-ordering this paragraph may help, because first you introduce channels for depolarization, then you go back to both de- and hyperpolarization. On the same note, which channels can be used for hyperpolarization of cardiomyocytes? I am not aware of any, even WiChR shows depolarizing effects in cardiomyocytes during prolonged activation (Vierock et al. 2022). Please delete: 'through a direct pathway' (Channelrhodopsins a directly light-gated channels, there are no pathways involved).*

We realised that the confusion arose from our use of incorrect terminology: we mistakenly wrote hyperpolarisation instead of repolarisation. In addition to channelrhodopsins such as WiChR, other tools can also induce a repolarising effect, including light-activatable chloride pumps (e.g., JAWS). However, to improve clarity, we recognize that repolarisation is not relevant to our manuscript and therefore decided to remove its mention (see below). Regarding the reported depolarising effects of WiChR in Vierock et al. (2022), we speculate that these may arise either from the specific phenotype of the cardiomyocytes used in the study, i.e. human induced pluripotent stem cell-derived atrial myocytes (aCMs), or from the particular ionic conditions applied during patch-clamp recordings (e.g., a bath solution containing 1 mM KCl). Notably, even after prolonged WiChR activation, the aCMs maintained a strongly negative maximum diastolic potential of approximately –55 mV.

"Although effects of illuminating miniSOG with light might lead to formation of depolarised areas, it is difficult to control the process precisely since it depolarises cardiomyocytes indirectly. Therefore, in this manuscript, we used light-sensitive ion channels to obtain more refined control over cardiomyocyte depolarisation. These ion channels allow the cells to respond to specific wavelengths of light, facilitating direct depolarisation (Ördög et al. (2021, 2023)). By inducing cardiomyocyte depolarisation only in the illuminated areas, optogenetics enables precise spatiotemporal control of cardiac excitability, an attribute we exploit in this manuscript (Appendix 2 Figure 1)."

> *(6) Figure 1: What would be the y-axis of the 'energy-like curves' in B? What exactly did you plot here?*

The graphs in Figure 1B are schematic representations intended to clarify the phenomenon for the reader. They do not depict actual data from any simulation or experiment. We clarified this misunderstanding by specifying that Figure 1B is a schematic representation of the effects at play in this paper.

"(B) Schematic representation showing how light intensity influences collective behaviour of excitable systems, transitioning between a stationary state (STA) at low illumination intensities and an oscillatory state (OSC) at high illumination intensities. Bi-stability occurs at intermediate light intensities, where transitions between states are dependent on periodic wave train properties. TR. OSC, transient oscillations."

To expand slightly beyond the paper: our schematic representation was inspired by a common visualization in dynamical systems used to illustrate bi-stability (for an example, see Fig. 3 in Schleimer, J. H., Hesse, J., Contreras, S. A., & Schreiber, S. (2021). Firing statistics in the bistable regime of neurons with homoclinic spike generation. Physical Review E, 103(1), 012407.). In this framework, the y-axis can indeed be interpreted as an energy landscape, which is related to a probability measure through the Boltzmann distribution:

$E = \beta \ln\left(\frac{1}{p}\right)$. Here, p denotes the probability of occupying a particular state (STA or OSC).

This probability can be estimated from the area (BCL × number of pulses) falling within each state, as shown in Fig. 4C. Since an attractor corresponds to a high-probability state, it naturally appears as a potential well in the landscape.

> *(7) Lines 92-93: 'this transition resulted for the interaction of an illuminated region with depolarized CM and an external wave train' - please consider rephrasing (it is not the region interacting with depolarized CM; and the external wave train could be explained more clearly).*

We rephrased our unclear sentence as follows:

“This transition resulted from the interaction of depolarized cardiomyocytes in an illuminated region with an external wave train not originating from within the illuminated region.”

> *(8) Figure 2 and elsewhere: When mentioning 'frequency', please state frequency values and not cycle lengths. Please also reconsider your distinction between high and low frequencies; 200 ms (5 Hz) is actually the normal heart rate for neonatal rats (300 bpm).*

In the revised version, we have clarified frequency values explicitly and included them alongside period values wherever frequency is mentioned, to avoid any ambiguity. We also emphasize that our use of "high" and "low" frequency is strictly a relative distinction within the context of our data, and not meant to imply a biological interpretation.

> *(9) Lines 129-131: Why not record optical maps? Voltage dynamics in the transition zone between depolarised and non-depolarised regions might be especially interesting to look at?*

We would like to clarify that optical maps were recorded for every experiment, and all experimental traces of cardiac monolayer activity were derived from these maps. We agree with the reviewer that the voltage dynamics in the transition zone are particularly interesting. However, we selected the data representations that, in our view, best highlight the main mechanisms. When we analysed full voltage profiles, they didn’t add extra insights to this main mechanism. As the other reviewer noted, the manuscript already presents a wide range of regimes, so we decided not to introduce further complexity.

> *(10) Lines 156-157: Why was the model not adapted to match the biophysical properties (e.g., kinetics, ion selectivity, light sensitivity) of Cheriff?*

The model was not adapted to the biophysical properties of Cheriff, because this would entail a whole new study involving extensive patch-clamping experiments, fitting, and calibration to model the correct properties of the ion channel. Beyond considerations of time efficiency, incorporating more specific modelling parameters would not change the essence of our findings. While numeric parameter ranges might shift, the core results would remain unchanged. This is a result of our experimental design where we applied constant illumination of long duration (6s or longer), thus making a difference in kinetical properties of an optogenetic tool irrelevant. In addition, we were able to observe qualitatively similar phenomena using many other depolarising optogenetic tools (e.g. ChR2, ReaChR, CatCh and more) in our *in-vitro experiments.* We ended up with Cheriff as our optotool-of-choice for the practical reasons of good light-sensitivity and a non-overlapping spectrum with our fluorescent dyes.

Therefore, computationally using a more general depolarising ion channel hints at the more general applicability of the observed phenomena, supporting our claim of a universal mechanism (demonstrated experimentally with CheRiff and computationally with ChR2).

> *(11) Line 158: 1.7124 mW/mm^2 - While I understand that this is the specific intensity used as input in the model, I am convinced that the model is not as accurate to predict behaviour at this specific intensity (4 digits after the comma), especially given that the model has not been adapted to Cheriff (probably more light sensitive than ChR2). Can this be rephrased?*

We did not aim for quantitative correspondence between the computational model and the biological experiments, but rather for qualitative agreement and mechanistic insight (see line 157). Qualitative comparisons are computationally obtained in a whole range of different intensities, as demonstrated in the 3D diagram of Fig. 4C. We wanted to demonstrate that at one fixed light intensity (chosen to be 1.7124 mW/mm^2 for the most clear effect), it was possible for all three states (STA, OSC. TR. OSC.) to coexist depending on the number of pulses and their period. Therefore the specific intensity used in the computational model is correct, and for reproducibility, we have left it unchanged while clarifying that it refers specifically to the *in silico* model:

“Simulating at a fixed constant illumination of 1.7124 mW/mm^2 and a fixed number of 4 pulses, frequency dependency of collective bi-stability was reproduced in Figure 4A.”

> *(12) Lines 160, 165, and elsewhere: 'Once again, Once more' - please delete or rephrase.*

We agree that we could have written these binding words better and reformulated them to:

“Similar to the experimental observations, only intermediate electrical pacing frequencies (500-ms period) caused transitions from collective stationary behaviour to collective oscillatory behaviour and ectopic pacemaker activity had periods (710 ms) that were different from the stimulation train period (500 ms). Figure 4B shows the accumulation of pulses necessary to invoke a transition from the collective stationary state to the collective oscillatory state at a fixed stimulation period (600 ms). Also in the in silico simulations, ectopic pacemaker activity had periods (750 ms) that were different from the stimulation train period (600 ms). Also for the transient oscillatory state, the simulations show frequency selectivity (Appendix 2 Figure 4B).”

> *(13) Line 171: 'illumination strength': please refer to 'light intensity'.*

We have revised our formulation to now refer specifically to “light intensity”:

“We previously identified three important parameters influencing such transitions: light intensity, number of pulses, and frequency of pulses.”

> *(14) Lines 187-188: 'the illuminated region settles into this period of sending out pulses' - please rephrase, the meaning is not clear.*

We reformulated our sentence to make its content more clear to the reader:

“For the conditions that resulted in stable oscillations, the green vertical lines in the middle and right slices represent the natural pacemaker frequency in the oscillatory state. After the transition from the stationary towards the oscillatory state, oscillatory pulses emerging from the illuminated region gradually dampen and stabilize at this period, corresponding to the natural pacemaker frequency."

> *(15) Figure 7: A)- please state in the legend which parameter is plotted on the y-axis (it is included in the main text, but should be provided here as well); C) The numbers provided in brackets are confusing. Why is (4) a high pulse number and (3) a low pulse number? Why not just state the number of pulses and add alpha, beta, gamma, and delta for the panels in brackets? I suggest providing the parameters (e.g., 800 ms cycle length, 2 pulses, etc) for all combinations, but not rate them with low, high, etc. (see also comment above).*

We appreciate the reviewer's comments and have revised the caption for figure 7, which now reads as follows:

"Figure 7. Phase plane projections of pulse-dependent collective state transitions. (A) Phase space trajectories (displayed in the Voltage – x_r plane) of the NRVM computational model show a limit cycle (OSC) that is not lying around a stable fixed point (STA). (B) Parameter space slice showing the relationship between stimulation period and number of pulses for a fixed illumination intensity (1.72 $mW/mm2$) and size of the illuminated area (67 pixels edge length). Letters correspond to the graphs shown in C. (C) Phase space trajectories for different combinations of stimulus train period and number of pulses (α: 800 ms cycle length + 2 pulses, β: 800 ms cycle length + 4 pulses, γ: 250 ms cycle length + 3 pulses, δ: 250 ms cycle length + 8 pulses). α and δ do not result in a transition from the resting state to ectopic pacemaker activity, as under these circumstances the system moves towards the stationary stable fixed point from outside and inside the stable limit cycle, respectively. However, for β and γ, the stable limit cycle is approached from outside and inside, respectively, and ectopic pacemaker activity is induced."

> *(16) Line 258: 'other dimensions by the electrotonic current' - not clear, please rephrase and explain.*

We realized that our explanation was somewhat convoluted and have therefore changed the text as follows:

"Rather than producing oscillations, the system returns to the stationary state along dimensions other than those shown in Figure 7C (Voltage and x_r), as evidenced by the phase space trajectory crossing itself. This return is mediated by the electrotonic current."

> *(17) Line 263: 'increased too much' – please rephrase using scientific terminology.*

We rephrased our sentence to:

"However, this is not a Hopf bifurcation, because in that case the system would not return to the stationary state when the number of pulses exceeds a critical threshold."

> *(18) Line 275: 'stronger diffusion/electrotonic influence from the non-illuminated region' - not sure diffusion is the correct term here. Please explain by taking into account the membrane potential. Please make sure to use proper terminology. The same applies to lines 281-282.*

We appreciate this comment, which prompted us to revisit on our text. We realised that some sections could be worded more clearly, and we also identified an error in the legend of Supplementary Figure 7. The corresponding corrections are provided below:

“However, repolarisation reserve does have an influence, prolonging the transition when it is reduced (Appendix 2 Figure 7). This effect can be observed either by moving further from the boundary of the illuminated region, where the electrotonic influence from the non-illuminated region is weaker, or by introducing ionic changes, such as a reduction in I_{Ks} and/or I_{to}. For example, because the electrotonic influence is weaker in the center of the illuminated region, the voltage there is not pulled down toward the resting membrane potential as quickly as in cells at the border of the illuminated zone.”

“To add a multicellular component to our single cell model we introduced a current that replicates the effect of cell coupling and its associated electrotonic influence.”

“Figure 7. The effect of ionic changes on the termination of pacemaker activity. The mechanism that moves the oscillating illuminated tissue back to the stationary state after high frequency pacing is dependent on the ionic properties of the tissue, i.e. lower repolarisation reserves (20% I_{Ks} + 50% I_{to}) are associated with longer transition times.”

> *(19) Line 289: -58 mV (to be corrected), -20 mV, and +50 mV - please justify the selection of parameters chosen. This also applies elsewhere- the selection of parameters seems quite arbitrary, please make sure the selection process is more transparent to the reader.*

Our choice of parameters was guided by the dynamical properties of the illuminated cells as well as by illustrative purposes. The value of –58 mV corresponds to the stimulation threshold of the model. The values of 50 mV and –20 mV match those used for single-cell stimulation (Figure 8C2, right panel), producing excitable and bistable dynamics, respectively. We refer to this point in line 288 with the phrase “building on this result.” To maintain conciseness, we did not elaborate on the underlying reasoning within the manuscript and instead reported only the results.

We also corrected the previously missed minus sign: -58 mV.

> *(20) Figure 8 and corresponding text: I don't understand what stimulation with a voltage means. Is this an externally applied electric field? Or did you inject a current necessary to change the membrane voltage by this value? Please explain.*

Stimulation with a specific voltage is a standard computational technique and can be likened to performing a voltage-clamp experiment on each individual cell. In this approach, the voltage of every cell in the tissue is briefly forced to a defined value.

> *(21) Figure 8C- panel 2: Traces at -20 mV and + 50 mV are identical. Is this correct? Please explain.*

Yes, that is correct. The cell responds similarly to a voltage stimulus of -20 mV or one of 50 mV, because both values are well above the excitation threshold of a cardiomyocyte.

> *(22) Line 344 and elsewhere: 'diffusion current' - This is probably not the correct terminology for gap-junction mediated currents. Please rephrase.*

A diffusion current is a mathematical formulation for a gap junction mediated current here, so , depending on the background of the reader, one of the terms might be used focusing on different aspects of the results. In a mathematical modelling context one often refers to a diffusion current because cardiomyocytes monolayers and tissues can be modelled using a reaction-diffusion equation. From the context of fine-grain biological and biophysical details, one uses the term gap-junction mediated current. Our choice is motivated by the main target audience we have in mind, namely interdisciplinary researchers with a core background in the mathematics/physics/computer science fields.

However, to not exclude our secondary target audience of biological and medical readers we now clarified the terminology, drawing the parallel between the different fields of study at line 79:

“These waves resulted from the interplay between the diffusion current (also known in biology/biophysics as the gap junction mediated current) and the bi-stable state that was induced in the illuminated region.”

> *(23) Lines 357-58: 'Such ectopic sources are typically initiated by high frequency pacing' - While this might be true during clinical testing, how would you explain this when not externally imposed? What could be biological high-frequency triggers?*

Biological high-frequency triggers could include sudden increases in heart rates, such as those induced by physical activity or emotional stress. Another possibility is the occurrence of paroxysmal atrial or ventricular fibrillation, which could then give rise to an ectopic source.

> *(24) Lines 419-420: 'large ionic cell currents and small repolarising coupling currents'. Are coupling currents actually small in comparison to cellular currents? Can you provide relative numbers (~ratio)?*

Coupling currents are indeed small compared to cellular currents. This can be inferred from the I-V curve shown in Figure 8C1, which dips below 0 and creates bi-stability only because of the small coupling current. If the coupling current were larger, the system would revert to a monostable regime. To make this more concrete, we have now provided the exact value of the coupling current used in Figure 8C1.

“Otherwise, if the hills and dips of the N-shaped steady-state IV curve were large (Figure 8C-1), they would have similar magnitudes as the large currents of fast ion channels, preventing the subtle interaction between these strong ionic cell currents and the small repolarising coupling currents (-0.103649 ≈ 0.1 pA).”

> *(25) Line 426: Please explain how ‘voltage shocks’ were modelled.*

We would like to refer the reviewer to our response to comment (20) regarding how we model voltage shocks. In the context of line 426, a typical voltage shock corresponds to a tissue-wide stimulus of 50 mV. Independent of our computational model, line 426 also cites other publications showing that, in clinical settings, high-voltage shocks are unable to terminate ectopic sustained activity, consistent with our findings.

> *(26) Lines 429 ff: 0.2pA/pF would correspond to 20 pA for a small cardiomyocyte of 100 pF, this current should be measurable using patch-clamp recordings.*

In trying to be succinct, we may have caused some confusion. The difference between the dips (≈-0.07 pA/pF) and hills (_≈_0.11 pA/pF) is approximately 0.18 pA/pF. For a small cardiomyocyte, this corresponds to deviations from zero of roughly ±10 pA. Considering that typical RMS noise levels in whole-cell patch-clamp recordings range from 2-10 pA , it is understandable that detecting these peaks and dips in an I-V curve (average current after holding a voltage for an extended period) is difficult. Achieving statistical significance would therefore require patching a large number of cells.

Given the already extensive scope of our manuscript in terms of techniques and concepts, we decided not to pursue these additional patch-clamp experiments.

> ***Reviewer #2 (Recommendations for the authors):***
>
> *Given the deluge of conditions to consider, there are several areas of improvement possible in communicating the authors' findings. I have the following suggestions to improve the manuscript.*
>
> *(1) Please change "pulse train" straight pink bar OR add stimulation marks (such as "*", or individual pulse icons) to provide better visual clarity that the applied stimuli are "short ON, long OFF" electrical pulses. I had significant initial difficulty understanding what the pulse bars represented in Figures 2, 3, 4A-B, etc. This may be partially because stimuli here could be either light (either continuous or pulsed) or electrical (likely pulsed only). To me, a solid & unbroken line intuitively denotes a continuous stimulation. I understand now that the pink bar represents the entire pulse-train duration, but I think readers would be better served with an improvement to this indicator in some fashion. For instance, the "phases" were much clearer in Figures 7C and 8D because of how colour was used on the Vm(t) traces. (How you implement this is up to you, though!)*

We have addressed the reviewer's concern and updated the figures by marking each external pulse with a small vertical line (see below).

> *(2) Please label the electrical stimulation location (akin to the labelled stimulation marker in circle 2 state in Figure 1A) in at least Figures 2 and 4A, and at most throughout the manuscript. It is unclear which "edge" or "pixel" the pulse-train is originating from, although I've assumed it's the left edge of the 2D tissue (both in vitro and silico). This would help readers compare the relative timing of dark blue vs. orange optical signal tracings and to understand how the activation wavefront transverses the tissue.*

We indicated the pacing electrode in the optical voltage recordings with a grey asterisk. For the *in silico* simulations, the electrode was assumed to be far away, and the excitation was modelled as a parallel wave originating from the top boundary, indicated with a grey zone.

> *(3) Given the prevalence of computational experiments in this study, I suggest considering making a straightforward video demonstrating basic examples of STA, OSC, and TR.OSC states. I believe that a video visualizing these states would be visually clarifying to and greatly appreciated by readers. Appendix 2 Figure 3 would be the no-motion visualization of the examples I'm thinking of (i.e., a corresponding stitched video could be generated for this). However, this video-generation comment is a suggestion and not a request.*

We have included a video showing all relevant states, which is now part of the Supplementary Material.

(4) Please fix several typos that I found in the manuscript:

(4A) Line 279: a comma is needed after i.e. when used in: "peculiar, i.e. a standard". However, this is possibly stylistic (discard suggestion if you are consistent in the manuscript).

(4B) Line 382: extra period before "(Figure 3C)".

(4C) Line 501: two periods at end of sentence "scientific purposes.." .

We would like to thank the reviewer for pointing out these typos. We have corrected them and conducted an additional check throughout the manuscript for minor errors.

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